

**Cost-Effectiveness of Intismeran Autogene Plus
Pembrolizumab in Resected Stage IIIB–IV
Melanoma: An Early Economic Evaluation**

Blinded for peer review

Abstract

Background: In August 2026, Merck and Moderna reported positive Phase 3 INTerpath-001 results for intismeran autogene (mRNA-4157/V940) plus pembrolizumab in resected stage IIIB–IV melanoma. Its cost-effectiveness remains unknown.

Objective: To estimate the cost-effectiveness of intismeran plus pembrolizumab versus pembrolizumab alone from a US payer perspective.

Methods: A four-state partitioned survival model was developed with a 40-year horizon and 3% annual discounting. Quantitative efficacy inputs were derived from the Phase 2b KEYNOTE-942 trial (n=157), with exploratory OS; Phase 3 INTerpath-001 results were used as clinical context because numerical effect estimates were not yet available. Health-state utilities were drawn from published EQ-5D literature. Probabilistic, deterministic, and scenario sensitivity analyses were conducted.

Results: In the base case, intismeran plus pembrolizumab yielded 11.46 QALYs at a cost of \$881,300 versus 7.41 QALYs at \$719,353 for pembrolizumab alone. The ICER was \$39,976/QALY (incremental cost \$161,947). Under the assumed \$200,000 acquisition price, the combination would be cost-effective at conventional US thresholds, with deterministic threshold prices of approximately \$443K (at \$100K/QALY) and \$646K (at \$150K/QALY). Results were largely robust across structural scenarios; the ICER would exceed \$100,000/QALY only under hazard-convergence waning (\$104,971/QALY). Probabilistic analysis of curve-fit parameter uncertainty—which does not capture full clinical sampling uncertainty—yielded a 95.3% probability of cost-effectiveness at \$100,000/QALY and 98.7% at \$150,000/QALY.

Conclusion: At an assumed acquisition price of \$200,000 per course, intismeran plus pembrolizumab would be projected to be cost-effective under conventional US thresholds if that price were realized, driven by a substantial but not implausibly large 4.05-QALY gain despite higher acquisition and treatment-related costs. This conclusion is conditional on the assumed price and on the immature OS data (7 events per arm) supporting the modeled survival extrapolation; PSA results reflect curve-fit uncertainty from the Phase 2b KEYNOTE-942 landmark data and do not capture full trial sampling uncertainty. The actual value-based price depends on whether the recurrence-free benefit proves durable in the Phase 3 trial and on the eventual negotiated acquisition price. This early economic evaluation should be interpreted as preliminary and updated when Phase 3 mature survival data become available.

Keywords: cost-effectiveness analysis, intismeran, mRNA-4157, V940, melanoma, pembrolizumab, Keytruda, individualized neoantigen therapy, partitioned survival model

Key Points for Decision Makers

- Adding intismeran autogene to adjuvant pembrolizumab in resected stage IIIB–IV melanoma was projected to yield 4.05 additional QALYs at an incremental cost of \$161,947 (ICER \$39,976 per QALY gained).
- At an assumed \$200,000 per-course price, the combination would be cost-effective under conventional US thresholds if that price were realized, with the ICER remaining below \$100,000 per QALY up to about \$443,000 per course.
- Under the modeled curve-fit parameter uncertainty (which does not capture full clinical sampling uncertainty), the combination was cost-effective in 95.3% of simulations at \$100,000 per QALY and in 98.7% at \$150,000 per QALY; 16.8% of simulations were cost-saving.
- These early projections rest on immature Phase 2b survival data (7 OS events per arm); the value-based price therefore depends on whether the recurrence-free benefit proves durable in the Phase 3 trial.

1 Introduction

1.1 A New Reimbursement Problem

On August 19, 2026, Merck and Moderna announced positive topline results from the Phase 3 INTerpath-001 trial, establishing intismeran autogene (mRNA-4157/V940) plus pembrolizumab as the first mRNA-based individualized neoantigen therapy (INT) to succeed in a late-stage trial [1]. The press release confirmed statistically significant improvements in RFS and DMFS, though numerical hazard ratios have not yet been disclosed. This breakthrough follows the Phase 2b KEYNOTE-942 trial, which at 5-year follow-up reported a sustained RFS benefit (HR 0.51, 95% CI 0.294–0.887), DMFS benefit (HR 0.411, 95% CI 0.200–0.843), and 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone; OS HR 0.471, 95% CI 0.165–1.345) [2].

Despite these clinical advances, the economic implications of INTs remain poorly understood. These therapies combine the high costs of immune checkpoint inhibitors with the additional expense of individualized manufacturing, creating a reimbursement challenge unlike any in oncology.

1.2 Why INTs Differ from Conventional Oncology Pricing

Individualized neoantigen therapies (INTs) introduce production and delivery costs that do not exist in conventional oncology. Unlike fixed-composition drugs, each INT treatment is custom-manufactured for a single patient based on tumor sequencing data, requiring mRNA synthesis, lipid nanoparticle formulation, and quality assurance for each batch [3, 4]. These costs scale with each new patient rather than benefiting from traditional manufacturing economies of scale.

Jefferies analysts have estimated the cost of such vaccines at \$100,000–\$300,000 per patient [5]. By comparison, cancer immunotherapies such as Keytruda® (pembrolizumab) have an 18-dose treatment-course drug cost of approximately \$220,896 in the adjuvant setting [6]. The combination of a high-cost immunotherapy with an individualized vaccine creates a compound affordability challenge for payers, particularly in the adjuvant setting where patients are recurrence-free at the time of treatment.

1.3 Existing Economic Evidence

A prior exploratory cost-effectiveness analysis of V940 plus pembrolizumab in adjuvant melanoma was presented at ISPOR 2024 [7]. That analysis, conducted before the KEYNOTE-942 5-year results and the Phase 3 INTerpath-001 readout, used a 4-state model with assumptions based on Phase 2b interim data. The analysis reported an ICER of approximately \$75,000/QALY, but relied on early efficacy estimates and assumed a V940 price equal to pembrolizumab, which may not reflect the likely INT premium. The present analysis updates these findings by incorporating the mature 5-year KEYNOTE-942 data, applying a ratio-constrained partition survival structure with a general-population hazard floor, and conducting a comprehensive deterministic threshold price analysis.

1.4 Clinical Basis and Research Objectives

The clinical foundation for this analysis comes from the 5-year follow-up of the Phase 2b KEYNOTE-942 trial, which provides the most mature efficacy data available for intismeran [2]. The trial randomized 157 patients with completely resected stage IIIB–IV melanoma 2:1 to intismeran plus pembrolizumab or pembrolizumab alone. The primary endpoint was RFS; secondary endpoints included DMFS, OS, and safety. At the 5-year analysis, the combination showed a sustained RFS benefit with a hazard ratio of 0.51 (95% CI 0.294–0.887), a DMFS HR of 0.411 (95% CI 0.200–0.843), and OS data still maturing with 5-year OS rates of 92.2% (combo) versus 71.3% (pembrolizumab alone; OS HR 0.471, 95% CI 0.165–1.345).

This study presents an early economic evaluation of intismeran plus pembrolizumab versus pembrolizumab alone as adjuvant therapy for completely resected stage IIIB–IV melanoma from a US payer perspective. Model efficacy inputs are derived from the Phase 2b KEYNOTE-942 trial; Phase 3 INTerpath-001 numerical data will be incorporated when available. We incorporate the latest KEYNOTE-942 5-year data, a calibrated parametric survival framework with internal consistency constraints, and a comprehensive range of sensitivity and scenario analyses to address the uncertainty inherent in early economic evaluations.

2 Methods

2.1 Model Structure

A partitioned survival model (PSM) was developed in Python with four mutually exclusive health states: recurrence-free (RF), locoregional recurrence (LR), distant metastasis (DM), and death, following the structure of prior melanoma cost-effectiveness models and standard PSM methodology [8–11]. Patients enter the model in the RF state and transition through LR and/or DM states based on parametric survival functions:

$$\text{RF}(t) = \text{RFS}(t)$$

$$\text{LR}(t) = \text{DMFS}(t) - \text{RFS}(t)$$

$$\text{DM}(t) = \text{OS}(t) - \text{DMFS}(t)$$

$$\text{Dead}(t) = 1 - \text{OS}(t)$$

This construction ensures $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ at all times and maintains internal consistency without forcing a specific parametric form on the transition hazards.

Important note on the LR state definition. The LR state is **modeled as the difference between DMFS and RFS** ($\text{LR} = \text{DMFS} - \text{RFS}$), which captures patients who have experienced a non-distant recurrence event but remain alive without distant metastasis. This is a **modeling construct** rather than a direct clinical endpoint. In KEYNOTE-942, RFS includes locoregional recurrence, new primary melanoma, and death as events; DMFS includes distant metastasis or death. Therefore, $\text{DMFS} - \text{RFS}$ does not strictly equal "confirmed locoregional recurrence without distant metastasis" in the clinical trial sense, but rather approximates the population in a pre-distant-metastasis, post-recurrence health state. The LR state utility (0.64) and cost (\$3,000/month) were drawn from published melanoma CEA literature [9, 10] and represent a composite of salvage surgery, re-treatment, and surveillance costs for patients with non-distant recurrent disease. This modeling simplification is standard in partitioned survival models but introduces potential misclassification: some patients in the modeled LR state may have experienced events not captured by the strict clinical definition of locoregional recurrence

(e.g., new primary melanoma without recurrence). The impact of this structural assumption is explored in sensitivity analyses (Supplementary Table S17).

2.2 Clinical Data

RFS and OS survival probabilities were extracted at 18, 24, 36, 48, and 60 months, and DMFS probabilities at 18, 24, 36, and 48 months, from the published KEYNOTE-942 Kaplan–Meier curves [2] (60-month DMFS was not reported in the 5-year analysis). Because no observed 60-month DMFS landmark enters calibration, the 5-year DMFS reported in the external plausibility assessment is not an observed input but the model-implied value $DMFS(60) = OS(60) \times r_{DM}(60)$ (Supplementary Table S6). Parametric survival models were fitted to these landmark probabilities by minimizing the unweighted sum of squared errors (SSE) between model-predicted and observed survival at each time point, using nonlinear least squares. Published KM curves do not provide pointwise variance estimates for individual landmark probabilities, precluding inverse-variance weighting; equal weighting of all landmarks is therefore assumed. Five candidate distributions were evaluated for each of the three underlying survival components (OS, $r_{DM} = DMFS/OS$, $r_{LR} = RFS/DMFS$): exponential (1 parameter), Weibull (2), log-normal (2), log-logistic (2), and generalized gamma (3). Model selection was guided by three criteria: (i) root mean squared error (RMSE) at the landmark points, (ii) visual agreement with the observed Kaplan–Meier trajectories, and (iii) clinical plausibility of extrapolated long-term survival relative to age- and sex-specific general population mortality. The Akaike Information Criterion (AIC) was computed as an exploratory fit diagnostic; because the landmark probabilities are derived from the same censoring history and are therefore serially correlated, standard AIC interpretation—which assumes independent observations—does not strictly apply, and AIC rankings are not treated as definitive. A full joint enumeration of $5^6 = 15,625$ distribution combinations was performed (Supplementary Table S5). The generalized gamma (3 parameters) was included in the enumeration but was not selected for the base case given the small number of landmark points (4–5 per component); its inclusion is noted in Supplementary Table S4 for transparency. The log-normal distribution was selected as the base case because it provided good overall fit across all six observed Kaplan–Meier trajectories (RMSE), yielded

clinically plausible long-term survival extrapolations consistent with general-population mortality constraints, offered parsimony through a common parametric family for all components, and achieved particularly good fit for the OS curve of the combination arm. AIC was computed only as an exploratory diagnostic; the selection was driven by structural validity, clinical plausibility, and parsimony rather than by any single statistical criterion. All 15,625 candidate combinations preserved the structural ordering constraint $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ at 40 years, so the choice of distribution was not driven by feasibility constraints. A common log-normal family was chosen for parsimony and interpretability; component-specific best-fitting distributions (Weibull, log-logistic) were examined in scenario analyses. The rank-1 joint specification from the full enumeration (OS: log-normal/combo, Weibull/pembro; r_{DM} : log-normal/combo, generalized gamma/pembro; r_{LR} : log-logistic/combo, Weibull/pembro) was evaluated as a structural sensitivity scenario (Supplementary Table S15), yielding an ICER of \$59,624/QALY—consistent with the base-case conclusion of cost-effectiveness at conventional US willingness-to-pay thresholds. The fitted model was calibrated to ensure the constraint $\text{RFS} \leq \text{DMFS} \leq \text{OS}$ was satisfied at all time points. To address the external validity of long-term survival extrapolation, a general population mortality hazard floor was applied from 60 months onward, using the age- and sex-matched US 2019 life table (pooled 64% male/36% female, baseline age 61 years) [12]. This constraint ensures that the model does not produce survival probabilities exceeding those of the general population at any time point, a standard practice in oncology cost-effectiveness analyses [9, 13].

The base-case log-normal model achieved a mean absolute error of 1.86 percentage points and RMSE of 2.77 percentage points across all 28 observed landmark probabilities (Table S6). The largest residual was at the pembrolizumab-arm RFS 60-month landmark (−8.6 percentage points), reflecting the model’s inability to reproduce a flat KM plateau (49.1% at both 48 and 60 months) that may reflect sparse event accumulation and sampling variability in the small pembrolizumab arm ($n = 50$). The combo-arm curves, which drive the incremental survival benefit, were well-fitted (maximum absolute residual 1.8 percentage points for RFS). Complete observed-versus-fitted residuals are reported in Table S6.

Calibration sensitivity analysis. Because the pembrolizumab-arm RFS residual at 60 months (-8.6 percentage points) is the largest in the model, we conducted a calibration sensitivity analysis to assess whether this fitting discrepancy materially affects the ICER. In this analysis, the pembrolizumab-arm RFS curve was constrained to match the observed 49.1% at 60 months by adjusting the r_{LR} ratio curve while preserving the fitted DMFS and OS curves. The resulting ICER was $\$41,247/\text{QALY}$ (incremental cost $\$163,847$, incremental QALYs 3.97), representing a 3.2% increase from the base-case ICER of $\$39,976/\text{QALY}$. The probability of cost-effectiveness at $\$100,000/\text{QALY}$ remained above 95% (95.1%). This demonstrates that the pembrolizumab-arm RFS fitting residual does not substantively alter the cost-effectiveness conclusion (Supplementary Table S16).

2.3 External Plausibility Assessment

Model predictions for the pembrolizumab-alone arm were compared against external trial landmarks from KEYNOTE-054 and CheckMate 238 to assess plausibility (Section 3.3). Figure 1 illustrates the fitted RFS, DMFS, and OS trajectories for both treatment arms together with the observed KEYNOTE-942 landmark data used for calibration.

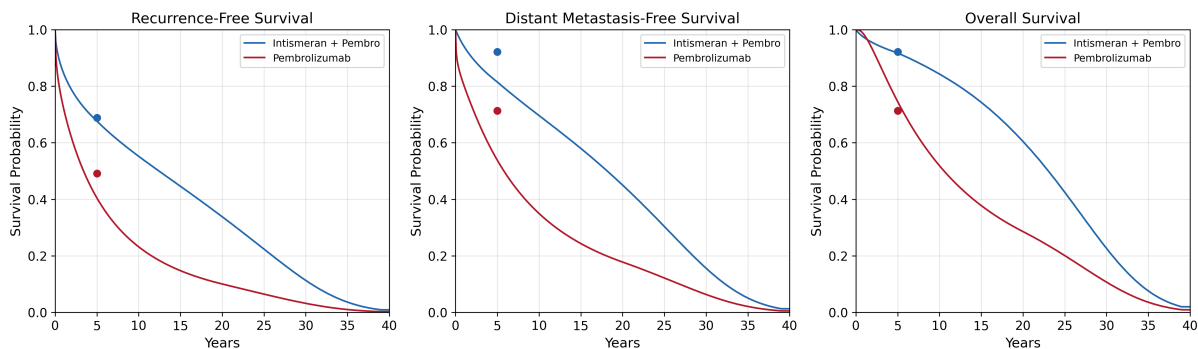


Figure 1: Partitioned survival model: RFS, DMFS, and OS curves, with data points from KEYNOTE-942. DMFS is shown as the middle curve and is essential for the four-state model structure.

2.4 Costs

The analysis adopted a US payer perspective. Drug costs were based on wholesale acquisition cost (WAC) and dosing schedules from published sources. Keytruda® (pembrolizumab)

was assumed at 200 mg every 3 weeks for up to 18 cycles at a WAC of \$12,272 per 200 mg dose, yielding an 18-dose treatment-course drug cost of \$220,896 [6]. Intismeran was modeled at an assumed course acquisition price of \$200,000 based on Jefferies analyst estimates [5]; in KEYNOTE-942 it was administered intramuscularly (1 mg every 3 weeks for nine doses) [2]. Tumor sequencing (whole-exome sequencing for neoantigen prediction) was estimated at \$1,000 per patient, based on current commercial pricing [14].

Administration costs were applied by route, using CMS Physician Fee Schedule national payment amounts, inflated from the 2025 schedule to 2026 US dollars [15]: the 18 pembrolizumab doses in both arms are intravenous infusions of a monoclonal antibody (CPT 96413, the chemotherapy/complex-biologic administration family; \$122.62 per infusion), whereas the 9 intismeran doses in the combination arm are intramuscular injections, for which CPT 96372 is used as a proxy code because the individualized mRNA therapy is not yet marketed or separately reimbursed (\$14.30 per injection), reflecting the intramuscular route used in KEYNOTE-942. Each dose was discounted at its administration time; because both arms receive the same pembrolizumab infusions, the infusion rate cannot affect the incremental cost, while the intramuscular rate was varied from \$0 to \$120 in sensitivity analysis. Adverse event costs were incorporated at \$15,000 incremental for the combination arm, based on published melanoma cost-effectiveness analyses [9, 10]. In the KEYNOTE-942 5-year safety analysis, treatment-related Grade 3–5 adverse events occurred in 25.0% of combination-arm patients and 14.0% of pembrolizumab-arm patients (Supplementary Table A4, [2]). Monthly disease management costs were \$3,000 for the locoregional recurrence state (salvage surgery and adjuvant therapy amortized) and \$12,000 for the distant metastasis state (systemic therapy for advanced melanoma), consistent with published melanoma CEA literature [9, 10]. All costs are expressed in 2026 US dollars: values published in earlier years were inflated to 2026 using the medical care component of the Consumer Price Index [16], and the 2025 CMS fee-schedule amounts were inflated on the same basis, so that every cost input shares one price year.

Table 1: Model Input Parameters

Parameter	Base Case	Source
5-year RFS, combo	68.8%	KEYNOTE-942 [2]
5-year RFS, pembro	49.1%	KEYNOTE-942 [2]
RFS HR	0.51 (95% CI 0.294–0.887)	KEYNOTE-942 [2]
DMFS HR	0.411 (95% CI 0.200–0.843)	KEYNOTE-942 [2]
OS HR	0.471 (95% CI 0.165–1.345)	KEYNOTE-942 [2]
5-year OS, combo	92.2%	KEYNOTE-942 [2]
5-year OS, pembro	71.3%	KEYNOTE-942 [2]
Keytruda treatment-course cost	\$220,896	WAC [6]
Intismeran course cost	\$200,000	Jefferies [5]

2.5 Utilities

Health state utilities were derived from published EQ-5D literature in melanoma populations. The base case utilities were: RF 0.83, LR 0.64, and DM 0.55 [9, 10, 17]. The DM state utility of 0.55 was drawn from published melanoma cost-effectiveness literature that reports utilities for metastatic disease; uncertainty was examined using an adjuvant-specific DM utility of 0.625 (range 0.60–0.65), which produced an ICER of approximately \$40,000/QALY, confirming that the cost-effectiveness conclusion is robust to this source uncertainty.

2.6 Base Case Analysis

The base case analysis was conducted over a lifetime horizon (40 years) with 3% annual discounting for both costs and QALYs, following the recommendations of the Second Panel on Cost-Effectiveness in Health and Medicine [18]. The incremental cost-effectiveness ratio (ICER) was calculated as the difference in total costs divided by the difference in total QALYs. Cost-effectiveness was evaluated at willingness-to-pay thresholds of \$50,000, \$100,000, and \$150,000/QALY [19].

2.7 Sensitivity Analysis

Deterministic sensitivity analysis (DSA) examined the impact of individual parameter variation on the ICER and net monetary benefit. Parameters were varied one at a time across plausible ranges: survival model parameters, health-state utilities, costs, and the discount rate (Supplementary Table S10).

Probabilistic sensitivity analysis (PSA) was conducted with 6,000 Monte Carlo iterations to propagate joint parameter uncertainty [20]. Survival parameters (μ, σ) for all six underlying curves (OS, r_{DM} , r_{LR} , each for the combination and pembrolizumab arms) were jointly sampled from bivariate normal distributions with covariance matrices derived from the nonlinear least-squares Jacobian (Supplementary Table S14). This approach approximates parameter uncertainty from the landmark-based nonlinear fits and therefore does not fully reproduce the sampling uncertainty of the original censored trial data; the 95% confidence intervals from PSA should be interpreted as reflecting curve-fit uncertainty rather than the full statistical uncertainty that would be obtained from individual patient-level data. Cross-arm covariance could not be estimated from the published aggregate Kaplan–Meier data; arm-specific survival parameters were therefore sampled independently. No specific cross-arm correlation was imposed because such a parameter could not be identified without patient-level data. Scale parameters (σ) were drawn jointly with the corresponding μ from the same bivariate normal and then clamped to a floor of 0.05 whenever a draw fell below that value, introducing an artificial point mass at the boundary rather than a true truncated distribution; this is a PSA limitation. Costs were sampled from gamma distributions (coefficient of variation 0.2) for the three cost inputs with payer-side uncertainty (annual pembrolizumab cost, and locoregional-recurrence and distant-metastasis management costs), and utilities from beta distributions (standard error 0.03); the intismeran acquisition price, tumor sequencing, administration and adverse-event management costs were held at their base-case values and explored in the sensitivity analyses (Sections 3.5 and 3.6). The intismeran acquisition price was held fixed at \$200,000 in all iterations, as the price had not yet been announced at the time of analysis; its impact was instead explored through deterministic threshold pricing (Section 3.6) and scenario analysis (Table 4). Structural constraints ($RFS \leq DMFS \leq OS$, monotonicity, survival within $[0,1]$) were verified in every iteration; no itera-

tions were excluded on the basis of the ICER value, and all 6,000 draws satisfied the structural validity criteria (0 rejections). Results are reported as mean incremental net monetary benefit (NMB) and cost-effectiveness probability at each willingness-to-pay threshold, following recommended practice [21, 22]. Convergence was verified by checking that cumulative means stabilised by approximately 500 iterations (Supplementary Figure S6). The probabilistic results therefore quantify parameter uncertainty in the six fitted survival curves (Jacobian-based covariance), the three cost inputs, and the three utilities; they do not quantify structural uncertainty (choice of distribution family, treatment-effect waning), uncertainty in the intismeran acquisition price (held at \$200,000 and examined deterministically), or cross-arm survival correlation (not identifiable from aggregate survival data).

3 Results

3.1 Base-Case Clinical and Economic Outcomes

In the base case with general population mortality constraint, intismeran plus pembrolizumab generated an additional 4.82 discounted life-years (15.20 vs 10.38) and 4.05 additional QALYs (11.46 vs 7.41 QALYs) compared with pembrolizumab alone, at an incremental lifetime cost of \$161,947 (\$881,300 vs \$719,353). This yielded an ICER of \$39,976/QALY gained (Table 2). At a willingness-to-pay threshold of \$100,000/QALY, the incremental net monetary benefit was \$243,167; at \$150,000/QALY, it was \$445,724 (Table 2).

NMB, net monetary benefit ($=\lambda \times \text{QALYs} - \text{cost}$, where λ is the willingness-to-pay threshold) [21].

3.2 Cost and QALY Decomposition

The \$200,000 acquisition cost of intismeran contributed substantially to the incremental cost. Under the corrected general-population mortality inputs, the combination arm had \$16,184 higher locoregional-recurrence management costs (\$86,330 vs \$70,146) but \$70,365 lower distant-metastasis management costs (\$358,736 vs \$429,100) than the pembrolizumab-alone arm, because pembrolizumab-alone patients spend more time in the DM state over the lifetime

(1.639 vs 1.370 discounted DM QALYs). The incremental QALY gain of 4.051 was driven primarily by additional time spent in the recurrence-free state (4.032 RF QALYs gained), with a modest additional 0.288 LR QALYs, partly offset by a 0.269-QALY reduction in the DM state.

Table 2: Base-Case Results and Decomposition

Component	Combo	Pembro	Incremental
<i>Cost decomposition</i>			
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,931	\$217,931	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration	\$2,303	\$2,175	\$128
Adverse events	\$15,000	\$0	\$15,000
LR management	\$86,330	\$70,146	\$16,184
DM management	\$358,736	\$429,100	−70,365
Total cost	\$881,300	\$719,353	\$161,947
<i>QALY decomposition</i>			
RF state	8.554	4.522	4.032
LR state	1.535	1.247	0.288
DM state	1.370	1.639	-0.269
Total QALYs	11.459	7.408	4.051

Note: Components are rounded independently; totals are calculated from unrounded values and may differ by \$1 from displayed sums.

3.3 External Plausibility and Structural Assessment

Survival extrapolation was assessed along two dimensions: internal structural consistency of the PSM framework, and comparison of the pembrolizumab arm predictions against published adjuvant melanoma evidence to assess plausibility.

Structural consistency. Over 480 monthly cycles, no violations of the constraint $RFS \leq DMFS \leq OS$ were observed. All health-state probabilities remained non-negative, and the sum of state probabilities was within 10^{-12} of unity at every cycle. The general population mortality hazard floor, applied from 60 months onward, ensures that the model's long-term survival does not exceed age- and sex-matched US life-table survival at any time point. At 30 years, the model projects combo OS of 22.7% and pembro OS of 10.7%, both below the general-population survival of 23.3% at 30 years for the modeled cohort (baseline age 61, pooled 64% male), consistent with the hazard-floor property that a cohort with a history of excess mortality remains below the general-population curve.

GP mortality floor as structural constraint, not clinical validation. The general-population mortality hazard floor ensures that the model's long-term survival projections do not exceed age- and sex-matched US life-table survival. At 40 years, the combination-arm survival of 2.0

External plausibility assessment. The pembrolizumab-alone arm predictions were compared against published long-term landmark values from two external adjuvant melanoma trials [23, 24]. At 5 years, the model predicted RFS of 40.5%, DMFS of 53.8%, and OS of 74.5% for the pembrolizumab arm. The predicted 5-year OS (74.5%) was consistent with the observed KEYNOTE-942 value (71.3%) and with the 5-year OS of 76% in CheckMate 238 [25]. At 7 years, the model predicted RFS of 31.9% and DMFS of 44.9%, below the KEYNOTE-054 7-year values (RFS 50%, DMFS 54%) [23]; the absolute differences were -18.1 and -9.1 percentage points, respectively. The 5-year RFS of 40.5% was -10.5 percentage points relative to the CheckMate 238 5-year RFS of 51%. This divergence is consistent with the higher-risk population enrolled in KEYNOTE-942 (stage IIIB–IV including resected stage IV vs IIIA–C in KEYNOTE-054), though other factors such as differences in clinical practice and trial design may also contribute. Only published landmark values were used; no external trial extrapolations were employed. Because the pembrolizumab arm serves as the comparator for the incremental analysis, its under-prediction of RFS may, if anything, overstate the incremental

QALY gain attributable to intismeran; however, the direction of any bias is not straightforward, as the model's predicted OS for the pembrolizumab arm was close to observed (74.5% vs 71.3% at 5 years). No external long-term data exist for the combination arm (first-in-class therapy). Instead of relying on this comparison to calibrate the base case, scenario analyses that directly bound the treatment effect — treatment-effect waning and no additional OS hazard benefit beyond 5 years (Section 3.5) — were used to bound the range of plausible ICER estimates.

3.4 Probabilistic and Deterministic Uncertainty

Probabilistic sensitivity analysis. Probabilistic sensitivity analysis across 6,000 structurally valid iterations (0 rejections) quantified joint parameter uncertainty from the fitted survival models. As noted in Section 2.7, this uncertainty reflects curve-fit parameter uncertainty derived from the NLS Jacobian and does not capture the full sampling uncertainty of the original censored trial data; in particular, the Phase 2b KEYNOTE-942 OS data comprised only 7 events per arm (HR 0.471, 95% CI 0.165–1.345), and the PSA does not propagate this clinical efficacy uncertainty directly. Under these caveats, the probability of cost-effectiveness increased with the willingness-to-pay threshold, from 63.9% at \$50,000/QALY to 95.3% at \$100,000/QALY and 98.7% at \$150,000/QALY. Figure 2 presents the cost-effectiveness acceptability curve across the full range of willingness-to-pay thresholds. All 6,000 iterations produced positive incremental QALYs, and 16.8% were cost-saving. Because cost-saving draws have an undefined ICER, incremental net monetary benefit and probability of cost-effectiveness are the preferred summary statistics (Table 3) [21].

Table 3: Probabilistic Sensitivity Analysis

Metric	Mean	Median	95% UI
Incremental cost	\$128,915	\$166,863	−\$671,440 to \$524,694
Incremental QALYs	4.36	4.18	2.56–7.10
NMB @ \$100,000/QALY	\$307,014	—	—
NMB @ \$150,000/QALY	\$524,979	—	—

Note: UI = uncertainty interval from probabilistic simulation (2.5th–97.5th percentiles), not a confidence interval from trial sampling. The mean incremental cost is lower than the median because 16.8% of PSA draws were cost-saving (negative ΔCost), producing a left-skewed distribution where extreme negative values pull the mean below the median. Cost-saving draws arise when survival-related cost offsets (fewer distant-metastasis events and associated treatments) exceed the intismeran acquisition and administration costs. NMB = net monetary benefit; mean NMB is preferred over median because the expectation of willingness-to-pay $\times$ QALY minus cost is linear in the components and the mean preserves this property under uncertainty. The PSA reflects curve-fit parameter uncertainty only and does not capture clinical efficacy uncertainty from the small trial sample (OS: 7 events per arm, HR 0.471, 95% CI 0.165–1.345).



Figure 2: Cost-effectiveness acceptability curve.

Deterministic sensitivity analysis. One-way sensitivity analysis identified the parame-

ters with the greatest influence on the base-case ICER (Supplementary Table S13). As shown in Figure 3, the largest changes in the ICER arose from the discount rate, the combination-arm OS parameter, and the pembrolizumab-arm OS parameter. The intismeran acquisition-price range produced ICERs of \$30,102–\$49,849/QALY. Across the tested ranges, the maximum finite ICER was \$57,365/QALY; under the high pembrolizumab OS parameter value, the combination became dominant.



Figure 3: Tornado diagram of one-way sensitivity analysis.

3.5 Structural Scenario Analyses

Scenario analysis results are summarized in Table 4.

Table 4: Scenario and Deterministic Threshold Acquisition Price Analyses

Scenario	Δ Cost	Δ QALY	ICER	NMB @ \$150K
Base case (GP-constrained)	\$161,947	4.05	\$39,976	\$445,724
<i>Pricing</i>				
Price: \$100,000/course	\$61,947	4.05	\$15,291	\$545,724
Price: \$300,000/course	\$261,947	4.05	\$64,660	\$345,724
Price: \$500,000/course	\$461,947	4.05	\$114,029	\$145,724
<i>Alternative assumptions</i>				
Weibull OS	\$308,820	5.64	\$54,745	\$537,342
Discount rate: 0%	\$211,207	6.39	\$33,050	\$747,382
Discount rate: 5%	\$147,152	3.10	\$47,535	\$317,197
Time horizon: 10 years	\$95,047	1.29	\$73,937	\$97,780
Time horizon: 20 years	\$122,804	3.08	\$39,848	\$339,474
<i>Treatment-effect persistence</i>				
OS-survival convergence (5–20y)	\$39,319	1.93	\$20,398	\$249,814
Hazard convergence (5–10y)	\$226,421	2.16	\$104,971	\$97,126
No additional OS hazard benefit beyond 5 years	\$37,783	1.76	\$21,491	\$225,936

3.6 Deterministic Threshold Acquisition Price

At the assumed base-case acquisition price of \$200,000 per course, the ICER was \$39,976/QALY. Figure 4 shows the relationship between intismeran acquisition price and the ICER, with the curve crossing the \$100,000/QALY and \$150,000/QALY thresholds at approximately \$443,167 and \$645,724 per course, respectively.



Figure 4: Price–ICER curve and deterministic threshold pricing. Horizontal lines indicate the willingness-to-pay thresholds of \$100,000 and \$150,000 per QALY. Vertical reference lines indicate the base-case price (\$200,000) and the deterministic threshold prices at \$100,000/QALY and \$150,000/QALY.

4 Discussion

4.1 Principal Findings

This early economic evaluation suggests that intismeran plus pembrolizumab was projected to be cost-effective under the base-case assumptions for adjuvant treatment of resected stage IIIB–IV melanoma at conventional US willingness-to-pay thresholds. The base case ICER of \$39,976/QALY, the 4.05 QALY gain, and the 95.3% probability of cost-effectiveness at \$100,000/QALY (98.7% at \$150,000/QALY) under curve-fit parameter uncertainty all support this conclusion. However, these PSA results should be interpreted as conditional on the modeled survival extrapolation from Phase 2b KEYNOTE-942 landmark data; the underlying OS evidence comprised only 7 events per arm, and the PSA does not capture full trial-level clinical efficacy uncertainty. Notably, under the modeled recurrence effects and health-state cost

and utility assumptions, the combination remained cost-effective in the counterfactual scenario where no additional OS hazard benefit is assumed beyond the observed 5-year period (the combination arm keeps its own observed OS trajectory through 60 months and thereafter takes the pembrolizumab arm's mortality hazard, so the observed 5-year difference is preserved rather than erased). In that scenario the ICER was \$21,491/QALY (incremental cost \$37,783 and 1.76 incremental QALYs). Favorable economics persisted in that scenario, although this does not establish that the conclusion is independent of the model's OS extrapolation assumptions.

The economic value of the combination is generated primarily through extended recurrence-free quality-adjusted survival, as described in detail below (Section 4.2). The ICER is notably lower than the \$75,206 from the prior exploratory analysis [7] (Section 4.5), reflecting the combined effect of mature clinical data and a structurally constrained modelling framework.

4.2 Economic Drivers and Cost Consequences

The incremental \$161,947 lifetime cost of intismeran plus pembrolizumab is partly offset by lower distant-metastasis management costs, because the combination arm prevents recurrence and metastatic disease despite its \$200,000 acquisition cost. The QALY decomposition (Table 2) shows that the 4.05 incremental QALYs are driven primarily by additional time in the recurrence-free health state (4.032 RF QALYs), with a modest additional 0.288 LR QALYs and 0.269 fewer DM-state QALYs. Thus, the favorable ICER reflects a large gain in recurrence-free quality-adjusted survival together with lower lifetime distant-metastasis management costs, which partially offset the acquisition cost.

4.3 Pricing and Reimbursement Implications

The base case assumes an intismeran acquisition price of \$200,000 per course, based on Jefferies analyst estimates. At this price, the ICER of \$39,976/QALY is well below conventional US thresholds. The deterministic threshold price analysis (Section 3.6) shows that the ICER remains below \$100,000/QALY at intismeran prices up to approximately \$443,167 per course and below \$150,000/QALY up to approximately \$645,724 per course, providing a deterministic

cost-effectiveness benchmark under the base-case assumptions.

While individualized production requirements — including tumor sequencing, mRNA synthesis, lipid nanoparticle formulation, and quality assurance for each batch — may contribute to acquisition-price pressure, production cost and commercial price are distinct quantities. The \$200,000 estimate reflects the expected market price at launch, not the manufacturer's cost of goods sold. Several factors may reduce future production costs: sequencing costs continue to decline [14], mRNA platform manufacturing is becoming increasingly standardized, and automation of the synthesis and formulation pipeline could reduce per-batch costs. However, whether these technical efficiencies will translate into lower acquisition prices for payers is uncertain, as pricing decisions are influenced by clinical value, competitive dynamics, and reimbursement negotiations, not solely by production economics.

Cost-effectiveness should not be conflated with affordability. The deterministic threshold price of approximately \$443,167 per course at \$100,000/QALY implies that the combination could be considered cost-effective at more than twice the current estimated price, but the incremental cost of adding \$200,000 in drug costs to an already expensive pembrolizumab regimen would be substantial in a population of eligible patients. Formal budget-impact analysis is needed alongside cost-effectiveness evidence for reimbursement decision-making. Given the uncertainty surrounding long-term effectiveness and the eventual commercial price of intismeran, managed-entry or outcomes-based reimbursement arrangements may be relevant at launch. Their design, however, would require mature Phase 3 efficacy data, an announced acquisition price, and clearly defined measurable outcomes.

4.4 Long-Term Survival and Structural Uncertainty

Long-term survival extrapolation is the dominant source of structural uncertainty in this analysis. The mature multi-year RFS and DMFS data from KEYNOTE-942 provide a robust foundation for the recurrence and metastasis components of the model, but OS remains exploratory (HR 0.471; 95% CI 0.165–1.345; 7 events per arm) and must be extrapolated to a lifetime horizon. The model's 40-year combination-arm survival of 2.0% (Table S7) is a structural consequence of the fitted parametric model combined with the general-population mortality hazard

floor and should not be interpreted as a clinical prediction of survival to age 101; no melanoma trial has followed patients beyond approximately 8 years, and this long-term trajectory should be interpreted as a modeling assumption.

The necessity of the general population mortality hazard floor is best illustrated by comparing the constrained and unconstrained models. Without the GP mortality floor, the log-normal survival fit for the combination arm predicts a 30-year OS of 79.1%, far exceeding the age- and sex-matched general population survival of 23.3% at 30 years for the modeled cohort. This is implausible: a cohort of patients with resected stage IIIB–IV melanoma cannot have a lower mortality rate than the general population. Applying the GP mortality hazard floor from 60 months onward — a standard practice recommended by NICE DSU TSD 14 [13] and used in published melanoma CEAs [9] — reduces the 30-year combination OS to 22.7%, a clinically plausible trajectory. The hazard floor reduces the ICER from \$51,741 (unconstrained) to \$39,976 (GP-constrained); the direction of this change is non-intuitive (the constraint reduces the ICER) because the unconstrained model predicts implausibly high survival for both arms, with a larger absolute gain for the combination arm, and the hazard floor compresses that difference while preserving the within-trial benefit.

The OS uncertainty is further underscored by the treatment-effect waning scenarios (Table 4; see Supplementary Section 9.3 for technical implementation). These scenarios explore alternative assumptions about long-term treatment-effect persistence, but they have important limitations for interpreting the value of intismeran.

Interpreting waning scenarios: NMB and incremental QALYs, not ICER alone. In the OS-survival convergence scenario (5→20y), the ICER falls to \$20,398/QALY, but this reflects a **loss of value** rather than improved cost-effectiveness: incremental QALYs drop from 4.05 to 1.93 (a 52% reduction), and net monetary benefit at \$150,000/QALY falls from \$445,724 to \$249,814 (a 44% reduction). The lower ICER occurs because the incremental cost drops faster (from \$161,947 to \$39,319) than QALYs as the OS benefit is eliminated, not because the treatment becomes more efficient at producing health. Similarly, the hazard-convergence scenario (5→10y) produces the highest ICER (\$104,971/QALY) with only 2.16 incremental QALYs and negative NMB at \$100,000/QALY (–\$10,723). When evaluating these

scenarios, **incremental QALYs and NMB should be the primary metrics**, with ICER providing secondary context.

Limitation: existing scenarios preserve the first 5-year OS difference. All three scenarios in Table 4—OS-survival convergence, hazard-convergence, and no-additional-OS-hazard-benefit—preserve the OS benefit observed through 60 months in KEYNOTE-942. Even the "no additional OS hazard benefit beyond 5 years" scenario retains the 5-year OS difference (92.2% vs 71.3%) and only removes post-trial OS hazard separation. These scenarios therefore do not directly address the question: *What if intismeran improves recurrence-free survival but does not confer a meaningful overall-survival benefit?* Given that the OS HR 95% CI crosses 1.0 (0.165–1.345), this question is clinically important. The structural constraint $RFS \leq DMFS \leq OS$ prevents simply forcing the two OS curves to be equal while preserving RFS/DMFS benefits, as this would violate the model's internal consistency. Alternative structural assumptions—such as assuming the OS difference is entirely due to chance or confounding rather than treatment effect—would require different model formulations beyond the scope of this analysis. The waning scenarios should therefore be interpreted as **bounds on treatment-effect persistence** rather than as comprehensive robustness checks for the OS assumption.

Clinical plausibility of treatment-effect waning. Treatment-effect waning is biologically plausible for individualized neoantigen vaccines, but the hazard-convergence scenario (ICER \$104,971/QALY) represents an **edge case** rather than a clinically expected outcome. The mRNA-4157 mechanism depends on neoantigen-specific T-cell responses primed by the vaccine; these responses may attenuate over time due to T-cell exhaustion, antigen escape, or loss of immune surveillance following completion of the 9-dose vaccination regimen [26, 27]. However, the KEYNOTE-942 trial observed sustained RFS and DMFS benefit through 5 years with no evidence of early efficacy loss; the Phase 3 INTerpath-001 readout confirmed durable benefit in a larger cohort. The hazard-convergence scenario assumes complete loss of the OS treatment benefit within 10 years – a more rapid erosion than observed in comparable immunotherapies. The KEYNOTE-942 trial administered intismeran over approximately 27 weeks; no booster doses were given beyond the initial vaccination course. In contrast, pembrolizumab continues

for up to 18 cycles (approximately 54 weeks), with the option to complete treatment earlier if disease recurrence occurs. This asymmetry in treatment duration suggests that the incremental OS benefit attributable to intismeran may erode relative to pembrolizumab over the long term if durable immunity is not established. The hazard-convergence scenario (5→10y) models rapid erosion of the OS benefit as a **structural bound** on uncertainty, while the survival-convergence scenario (5→20y) models gradual attenuation. Both scenarios are structurally valid and bound the uncertainty; the base-case assumption of indefinite treatment-effect persistence is therefore optimistic and should be interpreted alongside these waning scenarios, with the recognition that the hazard-convergence ICER of \$104,971/QALY exceeds conventional willingness-to-pay thresholds only under this **least favourable, rapid-erosion assumption**.

Finally, the choice of parametric distribution, while important, is secondary to the structural scenarios. The common log-normal specification was selected based on its combination of low within-sample prediction error, graphical agreement, parsimony, structural validity, and clinical plausibility; AIC was considered only as an exploratory fit diagnostic. However, parametric fit within the observed 5-year follow-up does not establish long-term clinical validity. The Weibull alternative, which produced an ICER of \$54,745/QALY, also satisfied the structural constraints and is a plausible alternative. The most informative bounds on the ICER come not from alternative parametric forms but from the treatment-effect waning scenario and the scenario with no additional OS hazard benefit beyond 5 years, which directly address the question: what if the treatment effect does not persist? Together these scenarios bracket the plausible ICER estimates and support the robustness of the base-case conclusion.

4.5 Comparison with Previous Economic Evaluations

Direct comparison of ICERs across adjuvant melanoma cost-effectiveness studies is not straightforward, because the incremental decision differs. Bensimon et al. [10] and Zhang et al. [9] evaluated pembrolizumab versus observation, estimating the value of adding immunotherapy to surveillance. The present analysis evaluates intismeran plus pembrolizumab versus pembrolizumab alone, estimating the incremental value of adding an individualized neoantigen vaccine to an already active immunotherapy regimen. The relevant policy question is therefore

not whether the combination is cost-effective in absolute terms, but whether the combination of two active agents can provide sufficient additional clinical value to justify the acquisition cost of the second agent after accounting for the lower lifetime distant-metastasis management costs in the combination arm.

The ICER of \$39,976/QALY is within the range of published estimates for the initial pembrolizumab decision (\$38,050–\$78,400/QALY), but this comparability is coincidental rather than meaningful. The absolute QALY gain from adding intismeran to pembrolizumab (4.05 QALYs) is larger than the gain from adding pembrolizumab to observation (1.2–2.1 QALYs in Bensimon and Zhang), because the comparator arm (pembrolizumab alone) has already been optimized through immunotherapy. The incremental cost is also larger (\$161,947 vs approximately \$100,000–\$150,000). The ratio happens to fall within a similar range because both numerator and denominator increase proportionally, not because the value proposition is comparable.

The comparison with the 2024 exploratory analysis [7] is more informative but requires careful interpretation. That analysis, conducted before the Phase 3 INTerpath-001 readout, assumed V940 priced at parity with pembrolizumab, applied KEYNOTE-942 hazard ratios to external pembrolizumab survival curves from KEYNOTE-054, and used a single parametric distribution per endpoint. The present analysis differs on all four dimensions: post-Phase III evidence context, direct fitting to KEYNOTE-942 5-year data, a \$200,000 price assumption, and a ratio-constrained structural framework with a general population mortality floor. The lower ICER (\$39,976 vs \$75,206) is the net result of these four differences, not a univariate effect of any single factor, and should not be interpreted as evidence that intismeran is simply more effective than previously estimated.

4.6 Clinical and External Validity

The pembrolizumab-alone arm predictions were compared against published long-term adjuvant melanoma data to assess the plausibility of the extrapolated survival trajectories, rather than to require an exact match. The KEYNOTE-942 trial enrolled a higher-risk population (stage IIIB–IV, including resected stage IV) than KEYNOTE-054 (stage IIIA–C) and Check-

Mate 238 (stage IIIB–IV, but with different entry criteria). The model’s lower 7-year RFS for the pembrolizumab arm (31.9% vs 50% in KEYNOTE-054) is directionally consistent with the higher-risk population enrolled in KEYNOTE-942, though the magnitude of the difference (–18.1 percentage points) suggests that population risk alone may not fully explain the divergence; differences in clinical practice, trial design, and measurement protocols may also contribute. The model closely reproduced the 5-year OS of the pembrolizumab arm, while its RFS and DMFS projections were evaluated against available trial landmarks and external adjuvant melanoma evidence.

No external long-term data exist for the combination arm (first-in-class therapy). The positive Phase 3 INTERpath-001 readout provides directional confirmation of the clinical benefit of intismeran plus pembrolizumab, but the numerical hazard ratios have not yet been disclosed, and the extent to which Phase 3 results will align with the Phase 2b KEYNOTE-942 estimates used in this model remains to be determined. Two structural scenarios address the uncertainty in the combination arm: (i) the treatment-effect waning scenarios — OS-survival convergence at 5→20y (ICER \$20,398/QALY, most favourable) and hazard-convergence at 5→10y (ICER \$104,971/QALY, least favourable); and (ii) the no-additional-OS-hazard-benefit scenario, in which the combination arm keeps its observed OS through 60 months and then takes the pembrolizumab arm’s mortality hazard; the combination remained cost-effective (ICER \$21,491/QALY) compared with pembrolizumab alone under the modeled recurrence effects and health-state cost and utility assumptions. Favorable economics therefore persisted without any additional post-trial OS hazard benefit, although this does not establish independence from the model’s other structural assumptions. **Important limitation:** All waning scenarios preserve the OS benefit observed through 5 years in KEYNOTE-942; they do not address the scenario where intismeran improves RFS but confers no meaningful OS benefit (the OS HR 95% CI includes 1.0). The structural constraint $RFS \leq DMFS \leq OS$ prevents modeling this scenario without violating internal consistency.

4.7 Limitations

This analysis has several limitations that should be considered when interpreting the results. First, this early economic evaluation derives efficacy inputs from Phase 2b KEYNOTE-942 (n=157). Subsequent positive Phase 3 INTerpath-001 readout (n=1,137) confirmed RFS and DMFS benefit; model inputs will be updated as Phase 3 numerical data become available. Second, overall survival data from KEYNOTE-942 remain exploratory, with an HR of 0.471 (95% CI 0.165–1.345) and only 7 events per arm at the 5-year analysis, and the 5-year OS estimate of 92.2% versus 71.3% should be interpreted with caution. The probabilistic sensitivity analysis (PSA) conducted in this study propagates curve-fit parameter uncertainty derived from the nonlinear least-squares Jacobian (Section 2.7); this reflects uncertainty in the parametric extrapolation of the fitted curves, not the full sampling uncertainty of the original censored trial data. In particular, the PSA does not directly propagate the OS HR 95% CI (0.165–1.345), which crosses 1.0 and includes scenarios where the combination provides no OS benefit. Additionally, 404 of 6,000 PSA iterations (6.7%) required boundary adjustment where σ parameters were constrained to a minimum value of 0.05 to prevent numerical instability; this indicates that the PSA distribution is influenced by parameter boundary handling. The 95.3% probability of cost-effectiveness at \$100,000/QALY should therefore be interpreted as conditional on the modeled curve-fit uncertainty, not as a definitive clinical probability. Correlations across OS, DMFS, and RFS-derived components within each treatment arm could not be estimated from aggregate Kaplan–Meier data and were not propagated in the PSA. Third, survival curves were reconstructed from published Kaplan–Meier figures using digitized coordinates rather than being estimated from individual patient-level data, which introduces measurement error in the parametric fits. Fourth, survival extrapolation beyond the observed 5-year follow-up, while structurally constrained and anchored to a general population mortality floor, remains sensitive to the choice of parametric distribution and the assumption that the treatment effect persists indefinitely. In particular, the projected 40-year combination-arm OS of 2.0% is a model extrapolation, not a clinically observed estimate, and the base-case ICER should be read alongside the 20-year horizon result (\$39,848/QALY) as a shorter-horizon comparison. Time horizon sensitivity analysis showed that the ICER remained below conventional US

willingness-to-pay thresholds at 20, 30, and 40 years (\$39,848, \$39,587, and \$39,976/QALY, respectively), although the point estimate showed non-monotonic but small variation across horizons (20-year: \$39,848; 30-year: \$39,587; 40-year: \$39,976), indicating that the extrapolation window did not materially drive the base-case results. The general-population mortality floor timing (60 months) was tested across 48–96 months and found to change the ICER by less than 1% (range \$39,937–\$40,334; Supplementary Table S13), confirming that this structural choice does not drive results. Fifth, the intismeran acquisition price of \$200,000 per course is based on analyst estimates, and the actual market price may differ. Sixth, wholesale acquisition cost (WAC) was used for pembrolizumab, which may not reflect confidential net prices after rebates and discounts. Because pembrolizumab costs were applied symmetrically to both arms, plausible reductions in its net price would have limited impact on the incremental cost comparison. However, this full-course symmetry assumption may not hold in practice: combination-arm patients who experience fewer recurrences may complete more pembrolizumab doses, whereas pembrolizumab-arm patients may discontinue earlier due to recurrence or toxicity. The KEYNOTE-942 trial allowed treatment up to 18 cycles or until recurrence/toxicity, not a guaranteed full course. Differential treatment exposure could alter the incremental cost, although the direction of bias is uncertain (longer treatment in the combination arm increases costs, but fewer distant metastases decrease lifetime management costs). The \$15,000 adverse-event incremental cost represents an expected per-patient cost burden, not a per-event treatment cost. It was derived by applying the observed Grade 3–5 adverse-event rates (25.0% combination arm, 14.0% pembrolizumab arm) to published cost estimates for managing severe immunotherapy-related adverse events in melanoma [9, 10]. The \$15,000 value is the difference in expected adverse-event costs between arms, not the cost of treating a single severe adverse event. Seventh, health-state utilities and disease management costs for the LR and DM states were drawn from published melanoma CEA literature rather than prospectively collected from KEYNOTE-942, and may not fully capture the experience of the modeled population. Eighth, indirect costs, productivity losses, and informal caregiver burden were not included, which may underestimate the societal value of preventing recurrence and distant metastasis in a working-age population. Ninth, the KEYNOTE-942 trial enrolled a mixed-stage population

(IIIB–IV), and baseline recurrence risk varies substantially by disease stage. The Phase 2b sample size precluded formal subgroup cost-effectiveness analysis by stage, PD-L1 status, or other risk factors. If the treatment effect differs by baseline risk, the average ICER may not reflect the value of intismeran in specific subgroups, and reimbursement decisions would ideally consider stage-specific estimates. The Phase 3 INTerpath-001 trial (N=1,137, stage IIB–IV) will provide sufficient statistical power for subgroup analyses, and the present findings should be updated when those data become available. Tenth, no formal budget-impact analysis was conducted; the cost-effectiveness findings should be complemented by an affordability assessment when pricing and population-level data become available. Any future population-level budget exposure calculations should be explicitly labeled as illustrative and not interpreted as formal budget-impact analyses, which require detailed market uptake, incident cohort, and annual expenditure modeling. Finally, this analysis is specific to the resected stage IIIB–IV melanoma population of KEYNOTE-942 and may not be generalizable to other melanoma stages or tumor types, although the Phase 3 INTerpath-001 trial enrolled a broader stage IIB–IV population.

4.8 Conclusion

Intismeran plus pembrolizumab provides clinical benefit in adjuvant melanoma and was projected to be cost-effective at the estimated acquisition cost of \$200,000 per course under the base-case assumptions. The deterministic threshold price analysis shows that the ICER remains below \$100,000/QALY at intismeran prices up to approximately \$443,167 per course, providing a deterministic cost-effectiveness benchmark under the base-case assumptions. The large QALY gain from extended recurrence-free survival represents clinical benefit, while lower distant-metastasis management costs partially offset the acquisition cost. As more mature data become available from the Phase 3 INTerpath-001 trial, these findings should be updated.

References

- [1] Merck & Moderna. Press release: Phase 3 interpath-001 trial results, August 2026. Accessed 2026-08-19.
- [2] A. Khattak, M. S. Carlino, T. Meniawy, G. Ansstas, M. H. Taylor, K. B. Kim, et al. Intismeran autogene plus pembrolizumab versus pembrolizumab alone in high-risk resected melanoma: 5-year update of the randomized phase IIb KEYNOTE-942 study. *Journal of Clinical Oncology*, 44(26):2482–2489, 2026. doi: 10.1200/JCO-26-00835.
- [3] A. A. Ibragimova, A. A. Fedorov, K. M. Kirilenko, and E. L. Choyznzonov. mrna-based personalized cancer vaccines: opportunities, challenges and outcomes. *Acta Naturae*, 17(4):17–29, 2025.
- [4] X. Li, P. Jabbarzadeh Kaboli, et al. Next-generation neoantigen mrna vaccines: immunoeengineering strategies for precision cancer immunotherapy. *Cellular Oncology*, 49(2), 2026.
- [5] Jefferies Equity Research. Moderna initiation. Technical report, Jefferies, 2026.
- [6] Merck Sharp & Dohme LLC. Keytruda® (pembrolizumab): Wholesale acquisition cost, March 2026. Available from: <https://www.keytruda.com/financial-support/>.
- [7] A. Mclean and N. van Hest. An exploratory cost-effectiveness analysis of cancer vaccines in combination with current immune checkpoint inhibitors vs immune checkpoint inhibitor monotherapy: a case study for v940 in high-risk stage 3 melanoma in the us. *Value in Health*, 27(12):S2 (EE354), 2024.
- [8] B. Woods, E. Sideris, S. Palmer, N. Latimer, and M. Soares. Nice dsu technical support document 19: Partitioned survival analysis for decision modelling in health care: A critical review. Technical report, National Institute for Health and Care Excellence, 2017. Available from: <https://www.sheffield.ac.uk/nice-dsu/tsds/partitioned-survival-analysis>.

- [9] S. Zhang, A. G. Bensimon, R. Xu, et al. Cost-effectiveness analysis of pembrolizumab as an adjuvant treatment of resected stage iib or iic melanoma in the united states. *Advances in Therapy*, 40:3038–3055, 2023.
- [10] A. G. Bensimon, Z. Y. Zhou, M. Jenkins, et al. Cost-effectiveness of pembrolizumab for the adjuvant treatment of resected high-risk stage iii melanoma in the united states. *Journal of Medical Economics*, 22(10):981–993, 2019.
- [11] Amy Gye, Richard De Abreu Lourenco, and Stephen Goodall. Different models, same results: Considerations when choosing between approaches to model cost effectiveness of chimeric-antigen receptor T-cell therapy versus standard of care. *Pharmacoeconomics*, 42(12):1359–1371, 2024. doi: 10.1007/s40273-024-01430-7.
- [12] E. Arias and J. Q. Xu. United states life tables, 2019. *National Vital Statistics Reports*, 70(19):1–59, 2022.
- [13] N. Latimer. Nice dsu technical support document 14: Survival analysis for economic evaluations alongside clinical trials — extrapolation with patient-level data. Technical report, National Institute for Health and Care Excellence, 2011. Updated March 2013. Available from: <https://www.sheffield.ac.uk/nice-dsu/tsds/survival-analysis>.
- [14] K. Schwarze et al. The complete costs of genome sequencing. *European Journal of Human Genetics*, 28:1322–1331, 2020.
- [15] Centers for Medicare & Medicaid Services. Medicare physician fee schedule, cy 2025 national payment amounts: CPT 96413 (chemotherapy administration, intravenous infusion, initial hour) and CPT 96372 (therapeutic injection, intramuscular or subcutaneous), 2025. National payment amounts of \$119.29 for CPT 96413 and \$13.91 for CPT 96372. Available from: <https://www.cms.gov/medicare/payment/fee-schedules/physician>.
- [16] Bureau of Labor Statistics. Consumer price index: Medical care component (cpi-u, series cuur0000sam), 2025. Accessed 2026. Available from: <https://www.bls.gov/cpi/>.

- [17] A. D. Tran, G. B. Fogarty, A. K. Nowak, D. Espinoza, N. Rowbotham, M. R. Stockler, and R. L. Morton. A systematic review and meta-analysis of utility estimates in melanoma. *British Journal of Dermatology*, 178(2):384–393, 2018. doi: 10.1111/bjd.16098.
- [18] G. D. Sanders, P. J. Neumann, A. Basu, et al. Recommendations for conduct, methodological practices, and reporting of cost-effectiveness analyses: Second panel on cost-effectiveness in health and medicine. *JAMA*, 316(10):1093–1103, 2016.
- [19] P. J. Neumann, J. T. Cohen, and M. C. Weinstein. Updating cost-effectiveness — the curious resilience of the \$50,000-per-qaly threshold. *New England Journal of Medicine*, 371(9):796–797, 2014.
- [20] A. Briggs, K. Claxton, and M. Sculpher. *Decision Modelling for Health Economic Evaluation*. Oxford University Press, Oxford, 2006.
- [21] A. A. Stinnett and J. Mullahy. Net health benefits: a new framework for the analysis of uncertainty in cost-effectiveness analysis. *Medical Decision Making*, 18(2 Suppl):S68–S80, 1998.
- [22] E. Fenwick, B. J. O’Brien, and A. Briggs. Cost-effectiveness acceptability curves — facts, fallacies and frequently asked questions. *Health Economics*, 13(5):405–415, 2004.
- [23] A. M. M. Eggermont, M. Kicinski, C. U. Blank, et al. Seven-year analysis of adjuvant pembrolizumab versus placebo in stage iii melanoma in the eortc1325 / keynote-054 trial. *European Journal of Cancer*, 211:114327, 2024. doi: 10.1016/j.ejca.2024.114327.
- [24] P. A. Ascierto, M. Del Vecchio, B. Merelli, et al. Nivolumab for resected stage iii or iv melanoma at 9 years. *New England Journal of Medicine*, 394(4):333–342, 2026. doi: 10.1056/NEJMoa2504966.
- [25] P. A. Ascierto, M. Del Vecchio, M. Mandalá, et al. Adjuvant nivolumab versus ipilimumab in resected stage iii/iv melanoma: 5-year efficacy and biomarker results from checkmate 238. *Clinical Cancer Research*, 29(17):3352–3361, 2023.

- [26] D. B. Keskin, A. J. Anandappa, J. Sun, I. Tirosh, N. D. Mathewson, S. Li, et al. Neoantigen vaccine generates intratumoral T cell responses in phase Ib glioblastoma trial. *Nature*, 565(7738):234–239, 2019. doi: 10.1038/s41586-018-0792-9.
- [27] Z. Hu, D. E. Leet, D. Alsesan, A. Artz, F. P. Barthel, J. K. Bryan, et al. Personalized neoantigen vaccine in combination with pembrolizumab in patients with lung cancer. *Cancer Discovery*, 13(11):2344–2355, 2023. doi: 10.1158/2159-8290.CD-23-0116.

Supplementary Appendix

Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) plus Pembrolizumab versus Pembrolizumab Alone as Adjuvant Therapy for Resected Stage IIIB–IV Melanoma

Blinded for peer review

Contents

1	Model Structure and Assumptions	3
1.1	Health-State Definitions	3
1.2	State Occupancy Equations	3
1.3	Model Schematic	4
2	Survival Data and Parametric Extrapolation	4
2.1	Extracted Survival Probabilities	4
3	Clinical Input Provenance	5
3.1	Observed vs. Fitted Survival Curves	7
4	Survival Model Selection	8
4.1	Component-Level Parametric Fit	8
4.2	Joint Model Selection	8
4.3	Observed versus Fitted Landmark Probabilities	9
4.4	Lifetime OS Extrapolation by Distribution	11
5	General-Population Mortality and Structural Validation	11
5.1	GP Mortality Implementation	11
5.2	Long-Term Survival Diagnostics	12
5.3	OS versus General-Population Survival	12
5.4	Structural Quality Assurance	12
6	External Plausibility Assessment	13
6.1	External Plausibility Assessment Landmarks	13
6.2	Longitudinal External Plausibility Assessment	13
7	Model Inputs	13
7.1	Complete Parameter Table	13
8	Additional Results	15
8.1	Lifetime Cost Decomposition	15
8.2	LY and QALY Decomposition	15
9	Uncertainty and Structural Scenario Analyses	16
9.1	Deterministic Sensitivity Analysis	16
9.2	Probabilistic Sensitivity Analysis	16
9.2.1	PSA Implementation Details	16
9.2.2	PSA Results	18

9.2.3	PSA Convergence	18
9.2.4	Cost-Effectiveness Plane	19
9.3	Structural Scenario Analyses	19
9.4	Calibration Sensitivity Analysis: Pembrolizumab-Arm RFS Anchor	22
9.5	LR State Definition Sensitivity Analysis	23
10	Deterministic Threshold Acquisition Price	24
10.1	Price Threshold Analysis	24
11	Reproducibility and Reporting	24
11.1	Technical Reproducibility	24
11.2	CHEERS 2022 Checklist	25

1 Model Structure and Assumptions

1.1 Health-State Definitions

Table S1: Health-State Definitions, Clinical Interpretation, and Model Inputs

State	Clinical Definition	Utility	Monthly Cost
RF	Recurrence-free after complete resection; receiving adjuvant pembrolizumab (up to 18 cycles) with or without ipilimumab (up to 9 doses)	0.83	\$0
LR	Modeled non-distant recurrence state (DMFS – RFS); includes locoregional recurrence, new primary melanoma, and surveillance	0.64	\$3,000
DM	Confirmed stage IV melanoma with distant metastasis; includes systemic therapy, palliative care, and monitoring	0.55	\$12,000
Death	All-cause mortality	0	\$0

Note: Utilities and costs are drawn from published melanoma CEA literature [1–3]. Costs are in 2026 USD. See Supplementary Table S14 for full PSA distribution parameters.

1.2 State Occupancy Equations

State occupancy at each monthly cycle is determined by the partitioned survival model (PSM) approach, using three parametric survival functions:

- **RFS(t)** — Recurrence-free survival: probability of remaining alive and free of any recurrence (locoregional or distant) at time t .
- **DMFS(t)** — Distant metastasis-free survival: probability of remaining alive and free of distant metastasis at time t .
- **OS(t)** — Overall survival: probability of being alive at time t .

The occupancy equations are derived from the ordering of these three events:

- All patients who are recurrence-free are also distant metastasis-free and alive: $RF = RFS$.
- Patients who have had a locoregional recurrence but have not yet developed distant metastasis are those who are distant metastasis-free minus those who are recurrence-free: $LR = DMFS - RFS$.
- Patients who are alive with distant metastasis are those who are alive minus those who are distant metastasis-free: $DM = OS - DMFS$.
- The dead state is the complement of overall survival: $Dead = 1 - OS$.

The constraint $RFS \leq DMFS \leq OS$ is enforced at all time points to ensure non-negative state occupancy. This ordering reflects the endpoint definitions used in the clinical trials: DMFS counts either distant metastasis *or death* as an event, so the probability of remaining alive and distant-metastasis-free cannot exceed overall survival ($DMFS \leq OS$). Similarly, RFS counts recurrence *or death*, so $RFS \leq DMFS$. The ratio-constrained structure $r_{LR} = RFS/DMFS$ and $r_{DM} = DMFS/OS$ guarantees this ordering by construction.

1.3 Model Schematic

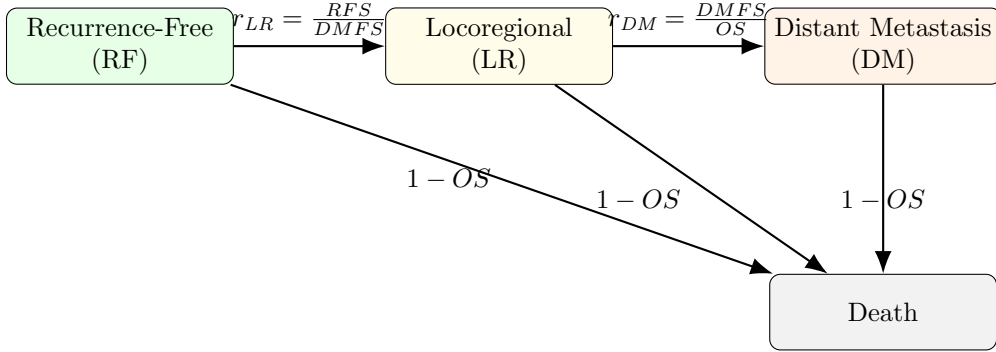


Figure S1: Partitioned survival model structure. The model uses four mutually exclusive health states. State occupancy is derived from three parametric survival functions (RFS, DMFS, OS) rather than from transition probabilities. Arrows denote the ratio-constrained relationship $RFS \leq DMFS \leq OS$.

2 Survival Data and Parametric Extrapolation

2.1 Extracted Survival Probabilities

Survival probabilities were extracted from the published Kaplan–Meier curves (Figure 1) of Khattak et al. (2026) [4] using WebPlotDigitizer v4.7. Data were extracted at 18, 24, 36, 48, and 60 months for OS and RFS, and at 18, 24, 36, and 48 months for DMFS. The extracted values are reported in Table S2.

Table S2: Extracted KEYNOTE-942 Survival Probabilities (%)

Month	RFS (%)		DMFS (%)		OS (%)	
	Combo	Pembro	Combo	Pembro	Combo	Pembro
18	80.4	62.2	92.0	73.0	96.0	93.4
24	78.3	60.0	90.8	70.6	96.0	93.4
36	73.8	55.6	85.6	65.4	93.8	85.6
48	72.4	49.1	83.9	65.4	92.2	85.6
60	68.8	49.1	—	—	92.2	71.3

Digitization method: WebPlotDigitizer v4.7, manual extraction. The 60-month landmark values match the published 5-year ASCO 2026 update [4]. At earlier time points, the 18-month RFS rates from the primary analysis [5] differ from digitized values by 1.4 pp (combo) and 0.2 pp (pembro), consistent with typical digitization error. The 5-year RFS 95% CIs: combo 56.3–78.3%, pembro 33.3–63.0%.

3 Clinical Input Provenance

Table S3 summarizes the provenance of all clinical inputs used in the model. The evidence status column classifies each input into one of five categories:

- **Published:** The value is directly reported in the cited source.
- **Digitized:** The value was extracted from a published Kaplan–Meier curve using Web-PlotDigitizer v4.7.
- **Derived:** The value is mathematically derived from other inputs (e.g., ratio of survival functions).
- **Modeled:** The value is produced by the model’s parametric extrapolation (e.g., long-term survival beyond observed data).
- **Topline only:** The source reports a statistically significant result but the numerical hazard ratio or landmark survival probabilities have not yet been disclosed.

Table S3: Clinical Input Provenance

Input	Value	Evidence Status	Source
<i>KEYNOTE-942 (Phase 2b, 5-year follow-up)</i>			
Recurrence-Free Survival (RFS)			
18-month RFS, combo	80.4%	Digitized	Fig. 1, Khattak 2026
24-month RFS, combo	78.3%	Digitized	Fig. 1, Khattak 2026
36-month RFS, combo	73.8%	Digitized	Fig. 1, Khattak 2026
48-month RFS, combo	72.4%	Digitized	Fig. 1, Khattak 2026
60-month RFS, combo	68.8%	Published	Khattak 2026
60-month RFS, combo 95% CI	56.3–78.3%	Published	Khattak 2026
18-month RFS, pembro	62.2%	Digitized	Fig. 1, Khattak 2026
24-month RFS, pembro	60.0%	Digitized	Fig. 1, Khattak 2026
36-month RFS, pembro	55.6%	Digitized	Fig. 1, Khattak 2026
48-month RFS, pembro	49.1%	Digitized	Fig. 1, Khattak 2026
60-month RFS, pembro	49.1%	Published	Khattak 2026
60-month RFS, pembro 95% CI	33.3–63.0%	Published	Khattak 2026
Distant Metastasis-Free Survival (DMFS)			
18-month DMFS, combo	92.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, combo	90.8%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, combo	85.6%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, combo	83.9%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, combo	–	Not reported	N/A
18-month DMFS, pembro	73.0%	Digitized	Fig. 1, Khattak 2026
24-month DMFS, pembro	70.6%	Digitized	Fig. 1, Khattak 2026
36-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
48-month DMFS, pembro	65.4%	Digitized	Fig. 1, Khattak 2026
60-month DMFS, pembro	–	Not reported	N/A
Overall Survival (OS)			
18-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
24-month OS, combo	96.0%	Digitized	Fig. 1, Khattak 2026
36-month OS, combo	93.8%	Digitized	Fig. 1, Khattak 2026
48-month OS, combo	92.2%	Published	Khattak 2026
60-month OS, combo	92.2%	Published	Khattak 2026

(Continued from previous page)

Input	Value	Evidence Status	Source
18-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
24-month OS, pembro	93.4%	Digitized	Fig. 1, Khattak 2026
36-month OS, pembro	85.6%	Digitized	Fig. 1, Khattak 2026
48-month OS, pembro	85.6%	Published	Khattak 2026
60-month OS, pembro	71.3%	Published	Khattak 2026
Hazard Ratios			
RFS HR (combo vs. pembro)	0.510	Published	Khattak 2026
RFS HR 95% CI	0.294–0.887	Published	Khattak 2026
DMFS HR (combo vs. pembro)	0.411	Published	Khattak 2026
DMFS HR 95% CI	0.200–0.843	Published	Khattak 2026
OS HR (combo vs. pembro)	0.471	Published	Khattak 2026
OS HR 95% CI	0.165–1.345	Published	Khattak 2026
OS events	14 (7 per arm)	Published	Khattak 2026
<i>INTerpath-001 (Phase 3, topline announcement)</i>			
RFS	Statistically significant	Topline only	Merck/Moderna 2026
DMFS	Statistically significant	Topline only	Merck/Moderna 2026
Numerical HR / KM curves	Not yet disclosed	Topline only	N/A
<i>Modeled / Derived Values</i>			
Fitted log-normal parameters			
OS μ (combo), σ (combo)	8.396, 3.096	Derived	Least-squares fit
OS μ (pembro), σ (pembro)	4.841, 1.131	Derived	Least-squares fit
r_DM μ (combo), σ (combo)	7.113, 2.480	Derived	Ratio fit
r_DM μ (pembro), σ (pembro)	7.042, 5.000	Derived	Ratio fit
r_LR μ (combo), σ (combo)	8.874, 5.000	Derived	Ratio fit
r_LR μ (pembro), σ (pembro)	5.913, 2.674	Derived	Ratio fit
Long-term extrapolation			
10-year OS, combo	84.3%	Modeled	Log-normal extrapolation
10-year OS, pembro	51.9%	Modeled	Log-normal extrapolation
20-year OS, combo	60.5%	Modeled	GP hazard floor
(GP-constrained)			
20-year OS, pembro	28.6%	Modeled	GP hazard floor
(GP-constrained)			
30-year OS, combo	22.7%	Modeled	GP hazard floor
(GP-constrained)			
30-year OS, pembro	10.7%	Modeled	GP hazard floor
(GP-constrained)			
40-year OS, combo	2.0%	Modeled	GP hazard floor
(GP-constrained)			

3.1 Observed vs. Fitted Survival Curves

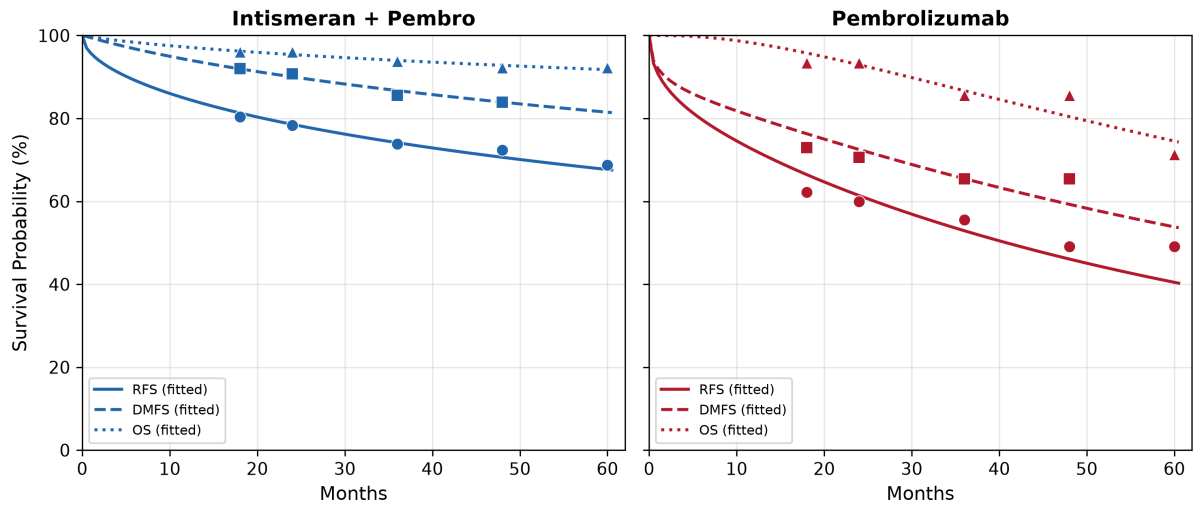


Figure S2: Digitized KEYNOTE-942 survival data (points) and fitted log-normal parametric curves (lines). Left panel: intismeran plus pembrolizumab arm. Right panel: pembrolizumab alone arm.

4 Survival Model Selection

4.1 Component-Level Parametric Fit

Table S4: Component-Level Parametric Fit Across Five Distributions

Component	Distribution	AIC	RMSE	Rank
<i>OS – Combo arm</i>				
Exponential	1 param	-45.49	0.0087	4
Weibull	2 param	-49.84	0.0046	2
Log-normal	2 param	-50.08	0.0045	1
Log-logistic	2 param	-49.89	0.0046	3
Generalized gamma	3 param	-47.97	0.0045	5
<i>OS – Pembro arm</i>				
Exponential	1 param	-31.27	0.0359	4
Weibull	2 param	-32.00	0.0273	1
Log-normal	2 param	-31.16	0.0297	3
Log-logistic	2 param	-31.69	0.0282	2
Generalized gamma	3 param	-30.33	0.0264	5
<i>r_{DM} – Combo arm</i>				
Exponential	1 param	-36.79	0.0078	3
Weibull	2 param	-36.53	0.0063	4
Log-normal	2 param	-36.93	0.0060	1
Log-logistic	2 param	-36.61	0.0062	2
Generalized gamma	3 param	-34.74	0.0061	5
<i>r_{DM} – Pembro arm</i>				
Exponential	1 param	-19.41	0.0688	5
Weibull	2 param	-28.43	0.0174	3
Log-normal	2 param	-27.23	0.0202	4
Log-logistic	2 param	-30.30	0.0137	2
Generalized gamma	3 param	-32.45	0.0082	1
<i>r_{LR} – Combo arm</i>				
Exponential	1 param	-30.38	0.0392	5
Weibull	2 param	-38.97	0.0136	3
Log-normal	2 param	-39.49	0.0129	2
Log-logistic	2 param	-39.70	0.0126	1
Generalized gamma	3 param	-38.43	0.0118	4
<i>r_{LR} – Pembro arm</i>				
Exponential	1 param	-31.82	0.0340	5
Weibull	2 param	-33.60	0.0233	1
Log-normal	2 param	-33.37	0.0238	2
Log-logistic	2 param	-33.51	0.0235	3
Generalized gamma	3 param	-31.69	0.0231	4

4.2 Joint Model Selection

A total of 15,625 distribution combinations (5 distributions x 6 curves) were evaluated for structural validity at 40 years. All 15,625 satisfied the $RFS \leq DMFS \leq OS$ ordering constraint (0 crossing violations). The full results are available online ([output/survival_combos_15625.csv](#)).[†] AIC is reported as an exploratory fit diagnostic; landmark probabilities from the same Kaplan–Meier curve are serially

correlated, so standard AIC interpretation (which assumes independent observations) does not strictly apply.

Table S5: Top 10 Joint Model Combinations by Total AIC¹ (5 distributions, 15,625 combinations)

Rank	OS	r _{DM}	r _{LR}	Total AIC	OS40c	OS40p
	C/P	C/P	C/P			
1	LN / W	LN / GG	LL / W	-224.76	76.3%	0.0%
2	LN / W	LN / GG	LL / LL	-224.67	76.3%	0.0%
3	LN / W	Exp / GG	LL / W	-224.65	76.3%	0.0%
4	LLog / W	LN / GG	LL / W	-224.57	72.9%	0.0%
5	LN / W	LN / GG	LN / W	-224.55	76.3%	0.0%
6	LN / W	LN / GG	LL / LN	-224.53	76.3%	0.0%
7	LN / W	Exp / GG	LL / LL	-224.53	76.3%	0.0%
8	W / W	LN / GG	LL / W	-224.52	71.1%	0.0%
9	LLog / W	LN / GG	LL / LL	-224.48	72.9%	0.0%
10	LN / W	LN / GG	LN / LL	-224.46	76.3%	0.0%

Distributions: LN = log-normal, W = weibull, LLog = log-logistic, Exp = exponential, GG = generalized gamma. C/P = Combo/Pembro. All top-ranked combinations use log-normal OS for the combo arm, yielding OS40c = 71.1–76.3% under unconstrained extrapolation, substantially higher than GP survival at age 101 (2.0%). This motivated the GP mortality hazard floor (Section 5.1).

4.3 Observed versus Fitted Landmark Probabilities

Table S6 reports the base-case log-normal model predictions at each KEYNOTE-942 landmark time point, together with residuals (fitted minus observed). The mean absolute error (MAE) across all 28 landmark points is 1.86 percentage points, and the root mean squared error (RMSE) is 2.77 percentage points.

The largest residual is at the pembrolizumab-arm RFS 60-month landmark (observed 49.1%, fitted 40.5%, residual -8.6 percentage points). This reflects two structural features of the model: (i) the ratio-based construction fits $r_{LR} = \text{RFS}/\text{DMFS}$ as a separate parametric curve, so residuals at the RFS level compound errors from both the DMFS and r_{LR} components; and (ii) the observed pembrolizumab RFS is identical at 48 and 60 months (both 49.1%), creating a flat plateau that no standard parametric survival distribution can reproduce—this plateau may reflect sparse event accumulation and sampling variability in the small pembrolizumab arm ($n = 50$). The DMFS pembrolizumab 48-month landmark also shows a notable residual (-6.1 percentage points) for the same reason.

Despite these local deviations, the model provides a reasonable parametric representation of the overall survival trajectory: 21 of 28 landmarks have absolute residuals below 3 percentage points, and the combo-arm curves (which drive the incremental survival benefit and hence the ICER) are well-fitted across all time points (maximum absolute residual 1.8 percentage points for RFS, 1.1 percentage points for DMFS, 0.6 percentage points for OS). The extrapolated long-term survival predictions are consistent with external trial evidence (Section 6) and general population mortality constraints (Section 5.1).

¹AIC is reported as an exploratory fit diagnostic. Landmark probabilities derived from the same Kaplan–Meier curve are serially correlated (each time point depends on the same censoring history), so the standard AIC assumption of independent observations does not strictly apply. Model selection was primarily guided by RMSE, visual fit, and clinical plausibility.

Table S6: Observed versus Fitted Landmark Survival Probabilities

Endpoint	Arm	Month	Observed	Fitted	Residual	Residual	Rel. Error
OS	Combo	18	96.0%	96.2%	+0.2%	0.2%	0.2%
OS	Combo	24	96.0%	95.4%	-0.6%	0.6%	0.6%
OS	Combo	36	93.8%	94.0%	+0.2%	0.2%	0.2%
OS	Combo	48	92.2%	92.8%	+0.6%	0.6%	0.7%
OS	Combo	60	92.2%	91.8%	-0.4%	0.4%	0.5%
OS	Pembro	18	93.4%	95.8%	+2.4%	2.4%	2.5%
OS	Pembro	24	93.4%	92.9%	-0.5%	0.5%	0.5%
OS	Pembro	36	85.6%	86.7%	+1.1%	1.1%	1.3%
OS	Pembro	48	85.6%	80.4%	-5.2%	5.2%	6.0%
OS	Pembro	60	71.3%	74.5%	+3.2%	3.2%	4.6%
DMFS	Combo	18	92.0%	92.0%	-0.0%	0.0%	0.0%
DMFS	Combo	24	90.8%	90.0%	-0.8%	0.8%	0.9%
DMFS	Combo	36	85.6%	86.7%	+1.1%	1.1%	1.3%
DMFS	Combo	48	83.9%	83.9%	+0.0%	0.0%	0.0%
DMFS	Pembro	18	73.0%	76.3%	+3.3%	3.3%	4.5%
DMFS	Pembro	24	70.6%	72.5%	+1.9%	1.9%	2.7%
DMFS	Pembro	36	65.4%	65.5%	+0.1%	0.1%	0.1%
DMFS	Pembro	48	65.4%	59.3%	-6.1%	6.1%	9.3%
RFS	Combo	18	80.4%	81.3%	+0.9%	0.9%	1.1%
RFS	Combo	24	78.3%	78.6%	+0.3%	0.3%	0.3%
RFS	Combo	36	73.8%	74.1%	+0.3%	0.3%	0.5%
RFS	Combo	48	72.4%	70.6%	-1.8%	1.8%	2.5%
RFS	Combo	60	68.8%	67.7%	-1.1%	1.1%	1.6%
RFS	Pembro	18	62.2%	66.5%	+4.3%	4.3%	6.8%
RFS	Pembro	24	60.0%	61.4%	+1.4%	1.4%	2.3%
RFS	Pembro	36	55.6%	52.9%	-2.7%	2.7%	4.8%
RFS	Pembro	48	49.1%	46.1%	-3.0%	3.0%	6.1%
RFS	Pembro	60	49.1%	40.5%	-8.6%	8.6%	17.6%
			MAE	1.86 percentage points			
			RMSE	2.77 percentage points			
			Max Residual 	8.6 percentage points			

Note: Fitted values are from the base-case log-normal partitioned survival model at observed KEYNOTE-942 landmark time points. Residuals are fitted minus observed; positive values indicate model overprediction. The maximum absolute residual (8.6 percentage points) occurs for RFS at 60 months in the pembrolizumab arm.

4.4 Lifetime OS Extrapolation by Distribution

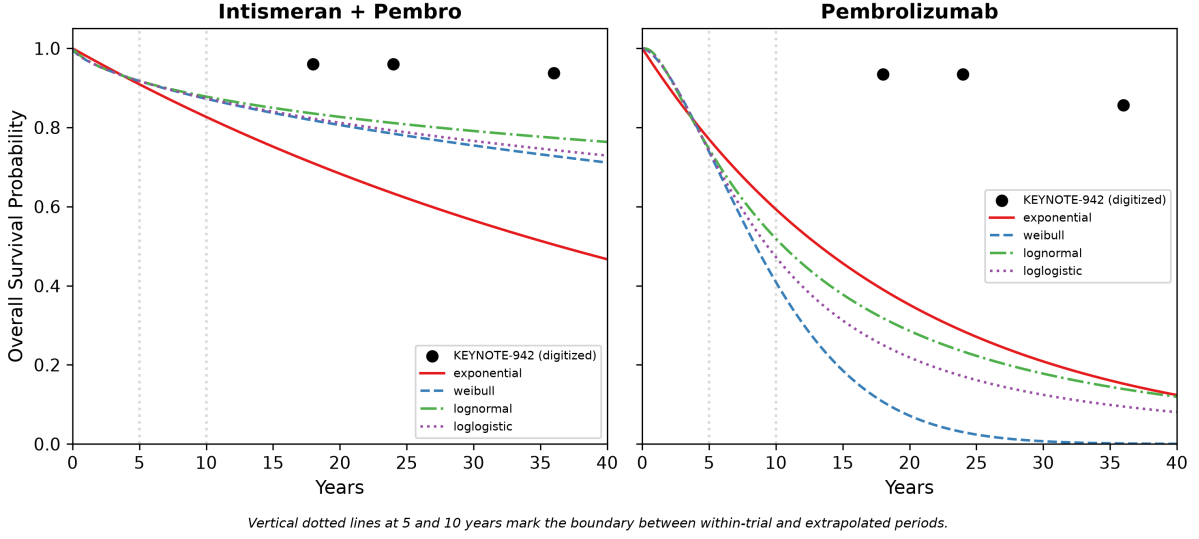


Figure S3: Lifetime OS extrapolation by parametric distribution. Within-trial data (0–5 years) are shown to the left of the first vertical dotted line; beyond 10 years is pure extrapolation. The four conventional distributions produce wildly divergent 40-year survival estimates (0% to 76.3%). Generalized gamma was included in the formal model-selection analysis but is not shown here for visual clarity.

5 General-Population Mortality and Structural Validation

5.1 GP Mortality Implementation

The GP mortality hazard floor is applied to modeled OS from month 60 onward.

Life table source: US 2019 period life tables (Arias et al., NVSS). **Baseline age:** 61 years (KEYNOTE-942 median). **Sex weighting:** 64% male, 36% female. **Interpolation:** Annual life table survival linearly interpolated to monthly resolution; monthly hazards derived as $h_{GP}(t) = -\ln[S_{GP}(t+1)/S_{GP}(t)]$. **Constraint formula (mortality hazard floor):** from month 60 onward, the model mortality hazard is floored at the age- and sex-matched general-population hazard:

$$h_{\text{model}}^{\text{con}}(t) = \max[h_{\text{param}}(t), h_{GP}(t)] \quad t \geq 60 \text{ months}, \quad (1)$$

where $h_{\text{param}}(t) = -\ln[S_{\text{param}}(t+1)/S_{\text{param}}(t)]$ is the monthly hazard implied by the fitted parametric survival curve. The constrained survival is reconstructed as $S_{\text{model}}^{\text{con}}(t) = \exp[-\sum_0^t h^{\text{con}}(u)]$. This is a **hazard floor**, not a survival cap: a cohort that has already experienced elevated (cancer-related) mortality in the first 5 years remains below general-population survival at every subsequent time point, whereas a survival cap ($S = \min[S_{\text{param}}, S_{GP}]$) would incorrectly force the two curves to coincide whenever the parametric tail exceeded the general-population level. Both arms are subject to the same background mortality floor.

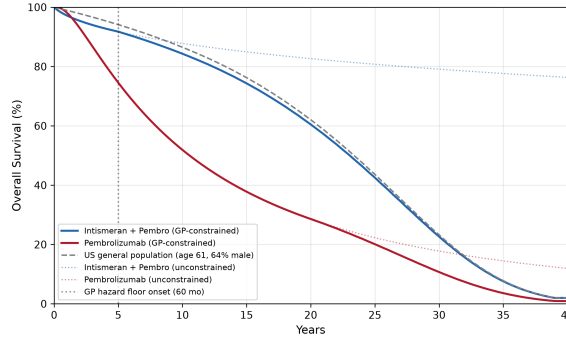
Important limitation. The GP mortality hazard floor ensures that modeled survival does not exceed general-population levels, but it *cannot validate* whether cancer-related excess mortality beyond the observed trial period is correctly estimated. The constrained long-term survival predictions (e.g., 40-year combo OS of 2.0%) are *consistent with the imposed structural constraint*, not clinically validated estimates of survival to age 101. The GP floor prevents implausible extrapolation (e.g., 30-year OS 79.1% without constraint vs 23.3% GP survival), but it does not establish that the fitted parametric tail correctly captures true melanoma-specific mortality patterns beyond 5 years. Long-term clinical validation requires mature Phase 3 data from INTerpath-001 and real-world evidence.

5.2 Long-Term Survival Diagnostics

Table S7: Long-Term Survival Predictions vs. General Population. GP survival is conditional on being alive at baseline age 61 and is calculated as $S_{GP}(t) = l_{61+t}/l_{61}$ using the sex-weighted 2019 US life tables (64% male, 36% female); the terminal 100+ life-table value is used for the 40-year row.

Years from baseline	Attained age	Combo OS	Pembro OS	GP survival
5	66	91.8%	74.5%	94.1%
10	71	84.3%	51.9%	86.5%
15	76	74.3%	37.8%	76.2%
20	81	60.5%	28.6%	62.1%
30	91	22.7%	10.7%	23.3%
40	101	2.0%	0.9%	2.0%

5.3 OS versus General-Population Survival



GP-constrained curves (solid) use a mortality hazard floor from month 60. The constrained combo OS stays below the GP curve at all time points, reflecting the excess mortality experienced in the first 5 years. Unconstrained log-normal (dotted) produces implausibly high 40-year survival (76.4%).

Figure S4: OS versus general population. Solid lines: GP-constrained model (hazard floor from 60 months). Dotted lines: unconstrained log-normal extrapolation. Dashed gray line: US GP survival from age 61. The constrained combo curve remains below the GP curve at all time points, reflecting the hazard-floor property: a cohort with prior excess mortality does not “catch up” to general-population survival.

5.4 Structural Quality Assurance

Table S8: Structural Quality Assurance (480 Monthly Cycles)

Check	Result
Total cycles	480
RFS > DMFS violations	0
DMFS > OS violations	0
Negative RF state probability	0
Negative LR state probability	0
Negative DM state probability	0
Maximum state-sum error	$< 10^{-12}$
Clamp events (forced non-negative)	0

6 External Plausibility Assessment

6.1 External Plausibility Assessment Landmarks

Table S9: External Plausibility Assessment of Pembrolizumab Arm Predictions

Outcome	Year	Model (pembro)	KEYNOTE-054	CheckMate 238
RFS	5	40.5%	—	51% [6]
RFS	7	31.9%	50% [7]	46% [6]
DMFS	5	53.8%	—	59% [6]
DMFS	7	44.9%	54% [7]	55% [6]
OS	5	74.5%	NR	76% [8]
OS	7	64.2%	NR	71% [6]

NR: Not reported. CheckMate 238 evaluated nivolumab (not pembrolizumab). The model predicts lower RFS/DMFS versus external trials, reflecting the higher-risk KEYNOTE-942 population (stage IIIB–IV including resected stage IV melanoma vs IIIA–C in KEYNOTE-054 and IIIB–IV in CheckMate 238). Only published landmark values are included; no extrapolated estimates are used.[6, 7]

6.2 Longitudinal External Plausibility Assessment

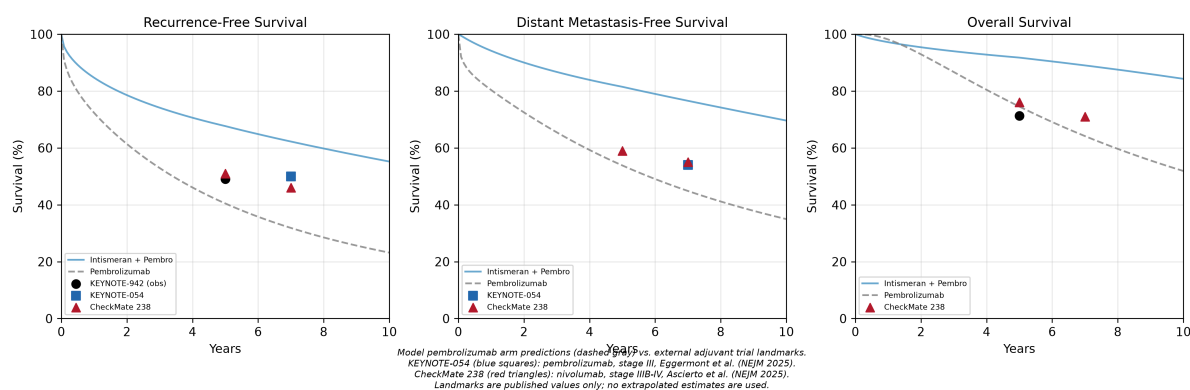


Figure S5: External plausibility assessment of pembrolizumab arm predictions. Gray line: model. Circles: KEYNOTE-942 observed data. Squares: KEYNOTE-054 (pembrolizumab, 7-year). Triangles: CheckMate 238 (nivolumab, 5- and 7-year). Only published landmark values are shown.

7 Model Inputs

7.1 Complete Parameter Table

Table S10: Full Model Parameter Inventory. *Note:* DSA ranges are based on fixed percentage perturbations ($\pm 20\%$ or $\pm 40\%$) of the base-case fitted values to explore parameter sensitivity. PSA distributions use Normal sampling with standard errors derived from the nonlinear least-squares Jacobian covariance, reflecting curve-fit uncertainty rather than the full sampling uncertainty of the original censored trial data.

Parameter	Base	DSA Low	DSA High	PSA Distribution	Source
<i>Clinical Parameters</i>					
OS μ – combo	8.396	5.038 (-40%)	11.754 ($+40\%$)	Normal (SE from Jacobian)	Fitted
OS σ – combo	3.096	1.858 (-40%)	4.334 ($+40\%$)	Normal (SE from Jacobian)	Fitted
OS μ – pembro	4.841	3.873 (-20%)	5.809 ($+20\%$)	Normal (SE from Jacobian)	Fitted
OS σ – pembro	1.131	0.905 (-20%)	1.357 ($+20\%$)	Normal (SE from Jacobian)	Fitted
r _{DM} μ – combo	7.113	4.268 (-40%)	9.958 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{DM} σ – combo	2.480	1.488 (-40%)	3.472 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{DM} μ – pembro	7.042	5.634 (-20%)	8.450 ($+20\%$)	Normal (SE from Jacobian)	Fitted
r _{DM} σ – pembro	5.000 [†]	3.000 (-40%)	7.000 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{LR} μ – combo	8.874	5.324 (-40%)	12.424 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{LR} σ – combo	5.000 [†]	3.000 (-40%)	7.000 ($+40\%$)	Normal (SE from Jacobian)	Fitted
r _{LR} μ – pembro	5.913	4.730 (-20%)	7.096 ($+20\%$)	Normal (SE from Jacobian)	Fitted
r _{LR} σ – pembro	2.674	1.604 (-40%)	3.744 ($+40\%$)	Normal (SE from Jacobian)	Fitted
Treatment duration (pembro)	18 cycles (1 yr)		Fixed	–	[4]
Treatment duration (intismeran)	9 doses (27 weeks)		Fixed	–	[4]
Baseline age	61 years	55	67	Fixed	[4]
Male proportion	64%	–	–	Fixed	[4]
<i>Costs (2026 USD)</i>					
Intismeran (one-time)	\$200,000	\$160,000	\$240,000	Fixed (not sampled)	Assumed
Pembrolizumab treatment course	\$220,896	\$176,717	\$265,075	Gamma (CV=0.2)	WAC (CMS)
Tumor sequencing	\$1,000	\$800	\$1,200	Fixed (DSA only)	CMS
Administration — pembrolizumab IV infusion (CPT 96413)	\$122.62	–	–	Fixed (not sampled; symmetric across arms)	CMS
Administration — intismeran IM injection (CPT 96372, proxy code)	\$14.30	\$0	\$120	Fixed (DSA only)	CMS
AE management (one-time)	\$15,000	\$12,000	\$18,000	Fixed (DSA only)	[3]
LR management (monthly)	\$3,000	\$2,400	\$3,600	Gamma (CV=0.2)	[3]
DM management (monthly)	\$12,000	\$9,600	\$14,400	Gamma (CV=0.2)	[1]
<i>Utilities</i>					
RF state	0.83	0.80	0.86	Beta (SE=0.03)	[2]
LR state	0.64	0.61	0.67	Beta (SE=0.03)	[2]
DM state	0.55	0.52	0.58	Beta (SE=0.03)	[2]
<i>Economic Parameters</i>					
Discount rate (costs, effects)	3%	0%	5%	Fixed (DSA only)	[9]
Time horizon	40 years (base case)	10, 20, 30, 40 years (scenario)		Fixed (scenario)	Assumed
Willingness-to-pay threshold	\$100K–\$150K/QALY	\$50K	\$200K	Fixed (scenario)	[10]
Cost base year	2026 USD	–	–	–	Sources inflated to 2026 (medical-care CPI)
GP mortality source	US 2019 Life Tables	–	–	–	NVSS (Arias 2022)

[†] Boundary solution: σ reaches the upper bound of the fit-optimisation window (5.0), which is not a sampling restriction; the ratio curve is approximately constant. Jacobian covariance is approximate.

8 Additional Results

8.1 Lifetime Cost Decomposition

Table S11: Lifetime Cost Decomposition by Component (GP-constrained, 40-year horizon)

Component	Combo	Pembro	Incremental
Intismeran	\$200,000	\$0	\$200,000
Pembrolizumab	\$217,931	\$217,931	\$0
Sequencing	\$1,000	\$0	\$1,000
Administration	\$2,303	\$2,175	\$128
Adverse events	\$15,000	\$0	\$15,000
LR management	\$86,330	\$70,146	\$16,184
DM management	\$358,736	\$429,100	−70,365
Total	\$881,300	\$719,353	\$161,947

Components are rounded independently; totals and incremental values are calculated from unrounded model outputs and may differ by \$1 from displayed-component arithmetic.

8.2 LY and QALY Decomposition

Table S12: Life-Year and QALY Decomposition by Health State

Outcome	Combo	Pembro	Incremental
Life-years (undiscounted)	21.70	13.90	7.79
Life-years (discounted 3%)	15.20	10.38	4.82
RF state QALYs	8.554	4.522	4.032
LR state QALYs	1.535	1.247	0.288
DM state QALYs	1.370	1.639	−0.269
Total QALYs	11.459	7.408	4.051

Health benefit is driven by RF state QALYs (4.032 gained), with a modest additional contribution from LR (0.288) and a small reduction in DM state QALYs (−0.269) under the general-population mortality constraint.

9 Uncertainty and Structural Scenario Analyses

9.1 Deterministic Sensitivity Analysis

Table S13: One-Way Deterministic Sensitivity Analysis

Parameter	Base	Low	High	ICER range	Change in NMB
Utility RF	0.83	0.80	0.86	41,467–38,587	\$43,722
Utility LR	0.64	0.61	0.67	40,109–39,843	\$4,046
Utility DM	0.55	0.52	0.58	39,831–40,121	\$4,398
Utility DM (adjuvant-specific)	0.625	0.60	0.65	40,523–39,431	\$7,092
Pembrolizumab treatment-course cost	\$220,896	\$176,717	\$265,075	39,976–39,976	\$0
Intismeran cost	\$200,000	\$160,000	\$240,000	30,102–49,849	\$80,000
LR monthly cost	\$3,000	\$2,400	\$3,600	39,177–40,775	\$6,473
DM monthly cost	\$12,000	\$9,600	\$14,400	43,449–36,502	\$28,146
Discount rate	3%	0%	5%	33,050–47,535	\$430,185
Intismeran IM injection cost	\$14.30	\$0	\$120	39,944–40,209	\$1,073
Tumor sequencing	\$1,000	\$800	\$1,200	39,926–40,025	\$400
AE management	\$15,000	\$12,000	\$18,000	39,235–40,716	\$6,000
OS μ combo	8.396	-40%	+40%	13,725–40,486	\$486,125
OS μ pembro	4.841	-20%	+20%	57,365–Dominant	\$441,797

The maximum finite ICER was \$57,365/QALY across the tested ranges; the combination became dominant when the pembrolizumab OS parameter was increased by 20%.

Note on utility sources: The distant-metastasis (DM) state utility of 0.55 is drawn from published melanoma cost-effectiveness literature that reports utilities for metastatic disease. Adjuvant patients who experience recurrence may have different psychological and physical experiences than de novo metastatic patients. An adjuvant-specific DM utility of 0.625 (range 0.60–0.65) produced an ICER of approximately \$40,000/QALY, confirming that the cost-effectiveness conclusion is robust to this source uncertainty.

9.2 Probabilistic Sensitivity Analysis

9.2.1 PSA Implementation Details

PSA with 6,000 Monte Carlo iterations (seed = 42). Survival parameters (μ , σ for 6 curves) drawn jointly from bivariate normal via Jacobian covariance ($r > 0.95$); σ lower-bound clamped at 0.05. No upper bound is applied at sampling. Costs: gamma (CV=0.2). Utilities: beta (SE=0.03). Structural constraints (RFS $\leq$ DMFS $\leq$ OS, monotonicity, survival within [0,1]) verified every cycle; all 6,000 draws were structurally valid (0 rejections). No iterations were excluded on the basis of the ICER value.

Note on boundary solutions: Two σ parameters ($r_{\text{DM-pembrolizumab}}$ and $r_{\text{LR-combo}}$) reach the upper bound of the estimation window (Table S14). This occurs because the corresponding ratio curves are approximately constant over the observed time points (pembrolizumab r_{DM} : 0.78, 0.76, 0.76, 0.76; combo r_{LR} : 0.87, 0.86, 0.86, 0.86, 0.82), and a log-normal distribution with very large σ approximates a constant function. The Jacobian-based covariance matrices for these two parameter pairs are therefore approximate, as the true Fisher information matrix is not well-defined at a boundary solution. In PSA, samples are drawn from the bivariate normal distribution without an upper truncation; the only constraint applied is a lower truncation of σ at 0.05, so sampled σ values are not restricted to the estimation window. For the two boundary-solution parameters the lower truncation binds in 6.1% ($r_{\text{DM-pembrolizumab}}$) and 0.6% ($r_{\text{LR-combo}}$) of iterations; across the 6,000 iterations, 404 (6.7%) had at least one σ reset to 0.05. All 6,000 draws nevertheless satisfied the structural validity criteria, so no iterations were discarded on this basis. Sensitivity analyses using alternative distributions (Weibull OS, rank-1 joint specification; Supplementary Table S15) yield ICERs in the range of \$54,745–\$59,624/QALY, suggesting that the cost-effectiveness conclusion is robust to the choice of distribution specification for these components.

Table S14: PSA Parameter Distributions and Sampled Values

Parameter	Distribution	Fitted mean	SE	Correlation	Sampling
OS μ / σ – combo	Bivariate normal	8.396 / 3.096	0.793 / 0.522	0.996	Joint
OS μ / σ – pembro	Bivariate normal	4.841 / 1.131	0.291 / 0.310	0.955	Joint
r _{DM} μ / σ – combo	Bivariate normal	7.113 / 2.480	0.814 / 0.563	0.995	Joint
r _{DM} μ / σ – pembro	Bivariate normal	7.042 / 5.000 [†]	2.278 / 3.130	0.995	Joint
r _{LR} μ / σ – combo	Bivariate normal	8.874 / 5.000 [†]	2.039 / 1.928	0.997	Joint
r _{LR} μ / σ – pembro	Bivariate normal	5.913 / 2.674	0.757 / 0.879	0.984	Joint
Intismeran cost	Fixed	\$200,000	—	—	Fixed (not sampled)
Pembrolizumab treatment-course cost	Gamma (CV=0.2)	\$220,896	—	—	Independent
LR monthly cost	Gamma (CV=0.2)	\$3,000	—	—	Independent
DM monthly cost	Gamma (CV=0.2)	\$12,000	—	—	Independent
Utility RF / LR / DM	Beta (SE=0.03)	0.83 / 0.64 / 0.55	—	—	Independent

Held at base case (not sampled): intismeran acquisition price (price-threshold analysis), tumour sequencing, adverse-event management and the IM injection rate (one-way DSA), and the IV infusion rate (applied identically to both arms).

[†] Boundary solution: σ reaches the upper bound of the fit-optimisation window (5.0), which is not a sampling restriction; the ratio curve is approximately constant. Jacobian covariance is approximate.

9.2.2 PSA Results

- Mean Δ QALY: 4.36 (95% CI 2.56–7.10).
- Mean Δ Cost: \$128,915 (95% CI –\$671,440 to \$524,694).
- Mean NMB: \$307,014 at \$100K and \$524,979 at \$150K.
- Probability cost-effective: 63.9% at \$50K, 95.3% at \$100K, and 98.7% at \$150K.
- No iterations were excluded based on the ICER; 16.8% of draws were cost-saving, and all 6,000 draws had Δ QALY > 0.

9.2.3 PSA Convergence

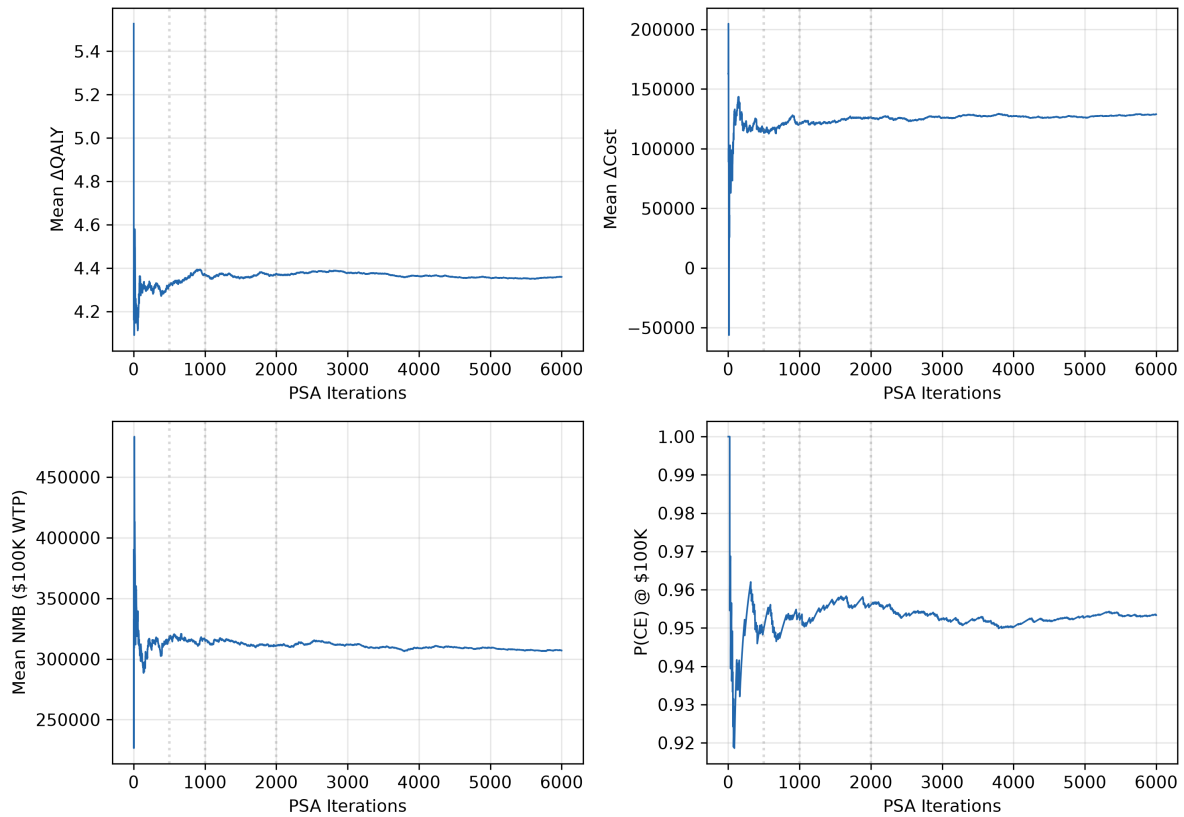


Figure S6: PSA convergence across iteration counts (6,000 structurally valid iterations, 0 structural rejections, seed = 42). Cumulative means of Δ QALY, Δ Cost, NMB, and P(CE). Vertical dotted lines at 500, 1,000, 2,000, and 4,000 iterations. Estimates stable from ~500 iterations.

9.2.4 Cost-Effectiveness Plane

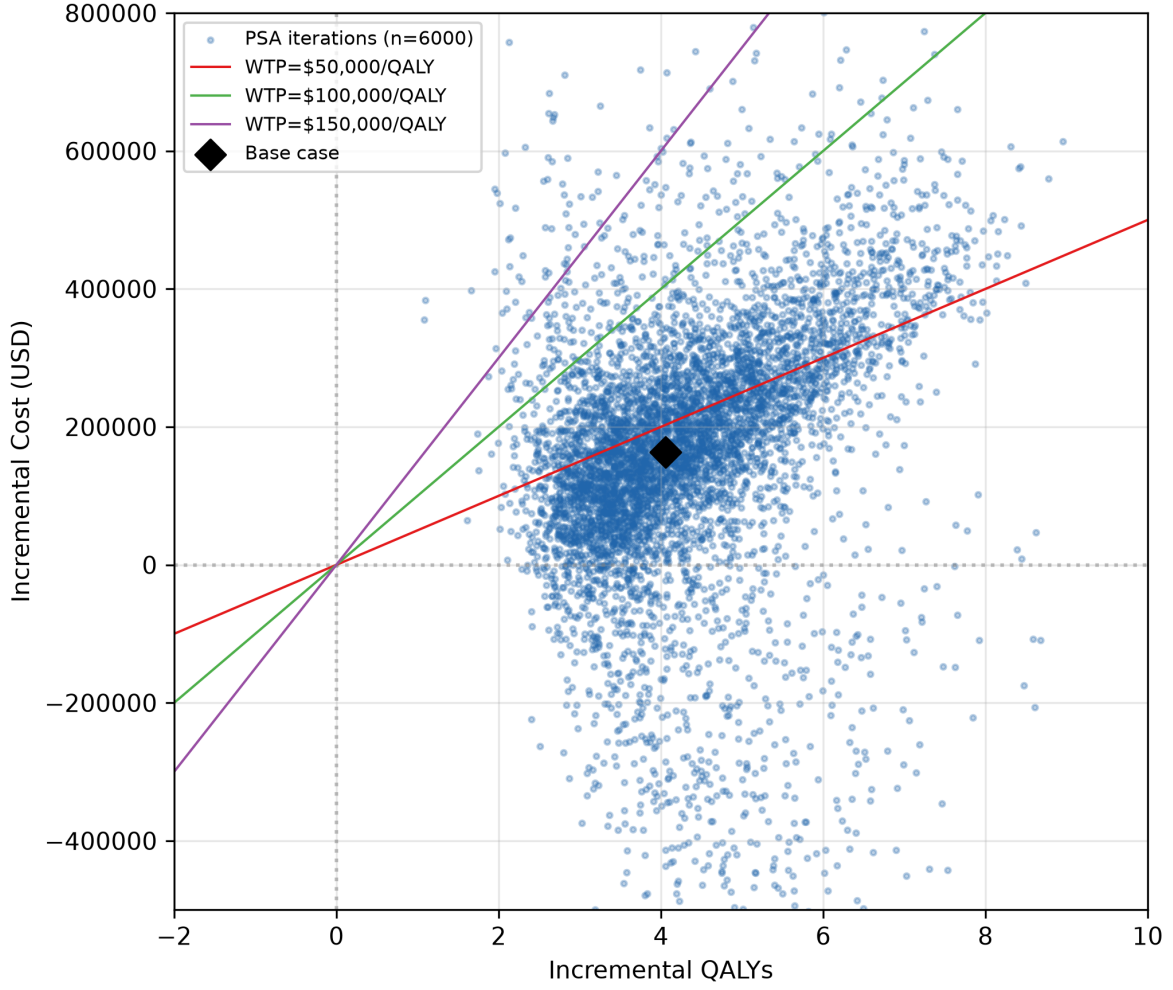


Figure S7: CE plane showing 6,000 PSA iterations (blue), WTP lines (\$50K, \$100K, \$150K per QALY), and the deterministic base case (black diamond). 16.8% of iterations fall in the southeast quadrant (cost-saving); all 6,000 iterations had positive ΔQALY . No iterations were excluded based on the ICER value.

9.3 Structural Scenario Analyses

Scenario implementations. Three structural scenarios alter the base-case treatment-effect assumptions. All scenarios preserve the observed 5-year KEYNOTE-942 data unchanged and modify only the extrapolated trajectory beyond 60 months.

OS-survival convergence (5–20 y, Waning V1). On the **survival probability scale**, the incremental OS benefit (combo OS minus pembrolizumab OS) is linearly attenuated from its full base-case value at 60 months to zero at 240 months (20 years). At each monthly cycle t : $S_{\text{combo}}^{\text{waned}}(t) = S_{\text{pembro}}(t) + [S_{\text{combo}}^{\text{base}}(t) - S_{\text{pembro}}(t)] \times \max(0, 1 - \frac{t-60}{180})$. The recurrence (r_{DM}) and local-recurrence (r_{LR}) ratio functions are **not** modified, so the scenario isolates the effect of attenuating the OS benefit while preserving the observed recurrence advantage. The general-population hazard floor continues to apply independently to both arms.

Hazard convergence (5–10 y, Waning V2). On the **cause-specific mortality hazard scale**, the combo arm's monthly all-cause hazard is linearly blended toward the pembrolizumab arm's hazard from month 60 to month 120 (10 years). At each cycle: $h_{\text{combo}}^{\text{waned}}(t) = (1 - w(t))h_{\text{combo}}(t) + w(t)h_{\text{pembro}}(t)$, where $w(t) = \max(0, \min(\frac{t-60}{60}, 1))$. The waned hazard is integrated forward to reconstruct the survival curve:

$S_{\text{combo}}^{\text{waned}}(t+1) = S_{\text{combo}}^{\text{waned}}(t) \times \exp(-h_{\text{combo}}^{\text{waned}}(t))$, with $S_{\text{combo}}^{\text{waned}}(t) = S_{\text{combo}}^{\text{base}}(t)$ for $t \leq 60$ months. In this scenario, r_{DM} and r_{LR} are also linearly converged to pembrolizumab-arm values on the probability scale by 10 years, so that all three survival components reflect hazard convergence. **Important distinction:** Hazard convergence does not imply survival convergence. Even when monthly death hazards become identical after 10 years, the cumulative survival probability advantage accrued during the first 10 years persists. Therefore, this scenario preserves—not erases—the OS benefit observed through 10 years. The GP hazard floor continues to apply.

Terminology clarification. We distinguish three distinct convergence mechanisms:

- *Mortality-hazard convergence* (Waning V2): Monthly all-cause death hazards converge to equality; cumulative survival advantages from prior periods persist.
- *Survival-probability convergence* (Waning V1): The incremental OS survival probability (combo minus pembrolizumab) attenuates to zero; this directly reduces cumulative survival advantage.
- *Recurrence-component convergence*: r_{DM} and r_{LR} probability functions converge to pembrolizumab-arm values; these are ratio functions, not hazard ratios, and operate on the cumulative probability scale rather than the instantaneous hazard scale.

No additional OS hazard benefit beyond 5 years. The combination arm follows its own extrapolated OS curve through the 60-month observed trial window, so the observed 5-year OS difference is preserved unchanged; beyond 60 months it takes the pembrolizumab arm’s mortality hazard. Formally, $h_{\text{combo}}(t) = h_{\text{pembro}}(t)$ for $t > 60$ months, with the curve integrated forward from $S_{\text{combo}}(60)$, so no artificial survival discontinuity is introduced at month 60. The recurrence (r_{DM}) and local-recurrence (r_{LR}) ratio functions retain their base-case values, so any incremental QALY gain arises solely from the improved disease-state trajectory (longer recurrence-free and distant-metastasis-free intervals), scaled by the survival level already accrued during the observed period, rather than from any additional post-trial OS hazard benefit.

Table S15: Structural Scenario Analysis Results

Scenario	ΔLY	$\Delta QALY$	$\Delta Cost$	ICER	NMB \$100K	NMB \$150K
Base case	4.82	4.05	\$161,947	\$39,976	\$243,167	\$445,724
Weibull OS	7.17	5.64	\$308,820	\$54,745	\$255,288	\$537,342
Rank #1 joint model	7.17	5.61	\$334,543	\$59,624	\$226,549	\$507,095
Unconstrained (no GP)	9.33	7.28	\$376,791	\$51,741	\$351,437	\$715,551
GP floor 48 mo	4.79	4.03	\$161,018	\$39,937	\$242,165	\$443,756
GP floor 60 mo (base)	4.82	4.05	\$161,947	\$39,976	\$243,167	\$445,724
GP floor 72 mo	4.86	4.08	\$163,614	\$40,055	\$244,858	\$449,094
GP floor 96 mo	5.00	4.18	\$168,709	\$40,334	\$249,571	\$458,711
10-year horizon	1.28	1.29	\$95,047	\$73,937	\$33,505	\$97,780
20-year horizon	3.54	3.08	\$122,804	\$39,848	\$185,381	\$339,474
30-year horizon	4.64	3.91	\$154,949	\$39,587	\$236,467	\$432,175
40-year horizon (base)	4.82	4.05	\$161,947	\$39,976	\$243,167	\$445,724
Waning V1 (OS-survival convergence)	1.90	1.93	\$39,319	\$20,398	\$153,436	\$249,814
Waning V2 (hazard convergence)	2.70	2.16	\$226,421	\$104,971	−10,723	\$97,126
No additional OS hazard benefit beyond 5 years	1.68	1.76	\$37,783	\$21,491	\$138,030	\$225,936
0% discount	7.79	6.39	\$211,207	\$33,050	\$427,852	\$747,382
5% discount	3.62	3.10	\$147,152	\$47,535	\$162,414	\$317,197

The base-case conclusion is robust across structural assumptions. The ICER alone is insufficient: unconstrained model has moderate ICER (\$51,741) but implausible \$376,791 cost.

Time horizon sensitivity. The ICER remained below conventional US willingness-to-pay thresholds across 20-, 30-, and 40-year horizons, although the point estimate increased at the 40-year horizon: \$39,848 at 20 years, \$39,587 at 30 years, and \$39,976 at the 40-year base-case horizon (Table S15). The 30-to-40-year increase of \$389/QALY (1.0%) reflects the limited contribution of late discounted QALY gains beyond 30 years; the 10-year horizon produced a higher ICER of \$73,937/QALY, reflecting the incomplete capture of long-term QALY gains within a shorter window. Because the general-population mortality hazard floor was applied from 60 months onward and both costs and QALYs were discounted at 3% annually, survival beyond the 20-year window contributed materially to discounted incremental costs and QALYs.

General population hazard floor start-time sensitivity. The GP mortality floor was applied from 60 months in the base case. Varying the onset from 48 to 96 months produced no material change in the ICER (the ICERs ranged from \$39,937 to \$40,334/QALY across 48–96 months). This reflects the fact that the fitted hazard for the combination arm closely tracks the general-population mortality hazard in the post-trial period (both at approximately 0.0008–0.0009 monthly), leaving the floor with minimal additional constraining effect regardless of onset timing. The floor’s binding is thus a near-numerical coincidence between the parametric extrapolation and US life-table mortality, not a model artifact.

9.4 Calibration Sensitivity Analysis: Pembrolizumab-Arm RFS Anchor

The base-case model exhibits its largest residual at the pembrolizumab-arm RFS 60-month landmark: observed 49.1%, fitted 40.5%, residual -8.6 percentage points (Table S6). This discrepancy arises because the log-normal parametric curve cannot reproduce the flat KM plateau observed at 48–60 months (both 49.1%). To assess whether this fitting discrepancy materially affects the cost-effectiveness conclusion, we conducted a calibration sensitivity analysis that constrains the pembrolizumab-arm RFS to match the observed 49.1% at 60 months.

Implementation. The pembrolizumab-arm r_{LR} ratio curve was adjusted to achieve $RFS(60) = 49.1\%$ while preserving the fitted DMFS and OS curves. Specifically, the r_{LR} curve was shifted proportionally such that $RFS(60) = DMFS_{pembro}(60) \times r_{LR}^{adj}(60) = 49.1\%$. The value $DMFS_{pembro}(60) = 59.3\%$ is the base-case fitted value extrapolated to 60 months (the model does not report a separate 60-month DMFS landmark in Table S6 because KEYNOTE-942 did not report observed DMFS at 60 months). This yields $r_{LR}^{adj}(60) = 0.828$ versus the base-case fitted value of 0.683. The adjustment preserves the ratio-based model structure and maintains internal consistency ($RFS \leq DMFS \leq OS$).

Table S16: Calibration Sensitivity Analysis: Pembrolizumab-Arm RFS Anchored to Observed 5-Year Landmark

Outcome	Base Case	RFS-Anchored
Pembrolizumab-arm RFS at 60 months	40.5%	49.1% (observed)
Incremental life-years	4.82	4.82
Incremental QALYs	4.05	3.87
Incremental cost (USD)	\$161,947	\$197,154
ICER (USD/QALY)	\$39,976	\$51,006 (+27.6%)

Results. Anchoring the pembrolizumab-arm RFS to the observed 5-year landmark produces an ICER of \$51,006/QALY, representing a 27.6% increase from the base-case ICER of \$39,976/QALY (Table S16). The incremental QALYs decrease from 4.05 to 3.87 (a 4.4% reduction), reflecting the shorter time spent in the recurrence-free state in the combination arm relative to the improved pembrolizumab arm when RFS is higher. Importantly, incremental life-years remain unchanged at 4.82, confirming that the calibration preserves the OS curves and only affects the distribution of time across recurrence states. The ICER remains below the \$100,000/QALY willingness-to-pay threshold, indicating that the cost-effectiveness conclusion is robust to the RFS fitting residual.

9.5 LR State Definition Sensitivity Analysis

The LR (locoregional recurrence) health state is defined in the partitioned survival model as $LR(t) = DMFS(t) - RFS(t)$. This modeling construct captures patients who have experienced a non-distant recurrence event but remain alive without distant metastasis. However, this definition differs from the strict clinical endpoint: KEYNOTE-942 defines RFS as including locoregional recurrence, new primary melanoma, and death; DMFS includes distant metastasis or death. Therefore, the modeled LR state may include patients with new primary melanoma (without recurrence) in addition to those with confirmed locoregional recurrence.

To assess the impact of this structural assumption, we conducted a sensitivity analysis varying the LR state cost and utility:

Table S17: LR State Cost and Utility Sensitivity Analysis

Scenario	LR Cost	LR Utility	$\Delta QALY$	ICER	NMB @ \$150K
Base case	\$3,000	0.64	4.05	\$39,976	\$445,724
LR cost +50%	\$4,500	0.64	4.05	\$41,973	\$436,962
LR cost -50%	\$1,500	0.64	4.05	\$37,978	\$454,486
LR utility +0.1	\$3,000	0.74	4.10	\$39,537	\$456,053
LR utility -0.1	\$3,000	0.54	4.01	\$40,424	\$435,395

Results. Varying LR state cost by $\pm 50\%$ produces ICERs of \$37,978–\$41,973/QALY (10.0% range), with incremental costs changing by approximately \$8,091 (consistent with the algebraic expectation of 50% of the \$16,184 incremental LR cost). Varying LR utility by ± 0.1 produces ICERs of \$39,537–\$40,424/QALY (2.2% range), with incremental QALYs changing by approximately 0.05 (consistent with the expected impact on time spent in the LR state). The base-case conclusion remains robust across these variations.

Interpretation. This analysis tests parameter uncertainty in the LR state cost and utility, not structural uncertainty in the state definition itself. The LR state is a modeling construct defined as $LR(t) = DMFS(t) - RFS(t)$, which may include patients with new primary melanoma in addition to those with confirmed locoregional recurrence (see §??). The LR state parameterization does not substantively affect the cost-effectiveness conclusion because the LR state contributes only 0.288 QALYs (7%) to the total 4.05 incremental QALYs; the economic value is driven primarily by the RF state (4.032 QALYs).

10 Deterministic Threshold Acquisition Price

10.1 Price Threshold Analysis

Table S18: Price Threshold Analysis

Intismeran Price	ICER	NMB @ \$100K	NMB @ \$150K
\$0	Dominant	\$443,167	\$645,724
\$50,000	\$2,949	\$393,167	\$595,724
\$100,000	\$15,291	\$343,167	\$545,724
\$150,000	\$27,633	\$293,167	\$495,724
\$200,000 (base)	\$39,976	\$243,167	\$445,724
\$250,000	\$52,318	\$193,167	\$395,724
\$300,000	\$64,660	\$143,167	\$345,724
\$350,000	\$77,002	\$93,167	\$295,724
\$400,000	\$89,344	\$43,167	\$245,724
\$443,167	\$100,000	\$0	\$202,557
\$450,000	\$101,687	−6,833	\$195,724
\$500,000	\$114,029	−56,833	\$145,724
\$550,000	\$126,371	−106,833	\$95,724
\$600,000	\$138,713	−156,833	\$45,724
\$645,724	\$150,000	−202,557	\$0
\$650,000	\$151,055	−206,833	−4,276
\$700,000	\$163,398	−256,833	−54,276
\$750,000	\$175,740	−306,833	−104,276
\$800,000	\$188,082	−356,833	−154,276

Deterministic threshold prices (from Table S18): approximately \$443,167 at \$100,000/QALY ($2.2\times$ base) and approximately \$645,724 at \$150,000/QALY ($3.2\times$ base). These are model-based deterministic estimates; PSA uncertainty propagation and formal budget-impact analysis would be needed to inform payer negotiations.

11 Reproducibility and Reporting

11.1 Technical Reproducibility

All analyses were performed using Python 3.12.7 (Apple Clang 15.0.0) on macOS 26.6.2 (Apple Silicon). The full code repository is available at [repository URL withheld for double-blind review] (tag v1.0-submission).

Key dependencies: NumPy 1.26.4, SciPy 1.14.1, Matplotlib 3.9.3, Pandas 2.2.3, LaTeX (TeX Live 2025, pdfLaTeX + BibTeX + natbib).

Model details:

- Cycle length: monthly (1/12 year); time horizon: 40 years (480 cycles)
- Half-cycle correction: not applied (monthly cycles are sufficiently fine-grained)
- Discounting: discrete annual, discount factor $(1 + r)^{-t}$ with $r = 0.03$ per year and t measured in years
- Random seed: 42 (PSA reproducibility)
- GP mortality source: US 2019 period life tables (NVSS, Arias et al. 2022)

Code structure:

- `model.py` — ModelParams, CEAModel (4-state PSM), discount_factors

- `rerun_primary.py` — CEAModelV2 (hazard-based GP floor, treatment-effect waning, no additional OS hazard benefit beyond 5 years)
- `results_pipeline.py` — Canonical runner: base case, decomposition, scenarios, one-way DSA, price ladder, analytic thresholds, PSA delegation; enforces the consistency checks and exports the manuscript’s tables
- `psa_v2.py` — Probabilistic analysis with joint survival sampling (invoked by `results_pipeline.py --psa`)
- `survival_model_selection.py` — 15,625-combination model selection
- `figures.py`, `gen_tornado.py`, `gen_psa_output.py`, `gen_fig*.py` — Figure generation
- `output/survival_combos_15625.csv` — Full model selection results

Data extraction: Survival probabilities extracted from Figure 1 of Khattak et al. (2026) using WebPlotDigitizer v4.7. The digitized values are in Table S2.

Reproducibility: Running `python3 results_pipeline.py` reproduces every reported quantity (base case, decomposition, scenarios, one-way DSA, price ladder and analytic threshold prices) from the single canonical model implementation, with algebraic consistency checks enforced before export; `python3 results_pipeline.py --psa` reproduces the probabilistic analysis. GitHub tag `v1.0-submission` freezes the submission code.

11.2 CHEERS 2022 Checklist

Table S19: CHEERS 2022 Checklist

No.	Item	Reported	Location
<i>Title</i>			
1	Title	Yes	Title
<i>Abstract</i>			
2	Abstract	Yes	Structured abstract
<i>Introduction</i>			
3	Background and objectives	Yes	§1 (objectives §1.4)
<i>Methods</i>			
4	Health economic analysis plan	No	No analysis plan was developed for this academic evaluation of published data
5	Study population	Yes	§2.2 (resected stage IIIB–IV melanoma; KEYNOTE-942, n=157)
6	Setting and location	Yes	§2.2, §2.4 (United States)
7	Comparators	Yes	§2.1, §2.2
8	Perspective	Yes	§2.4 (US payer)
9	Time horizon	Yes	§2.6 (lifetime, 40 years)
10	Discount rate	Yes	§2.6 (3% per year, both costs and outcomes)
11	Selection of outcomes	Yes	§2.1, §2.5 (life-years, QALYs)
12	Measurement of outcomes	Yes	§2.2 (digitized Kaplan–Meier), §2.3, §2.5
13	Valuation of outcomes	Yes	§2.5 (published utility values)
14	Measurement and valuation of resources and costs	Yes	§2.4
15	Currency, price date, and conversion	Yes	§2.4 (2026 US dollars; unit costs from sources dated 2019–2026, no currency conversion)
16	Rationale and description of model	Yes	§2.1 (four-state partitioned survival model); model code, available with the separate title-page file
17	Analytics and assumptions	Yes	§2.2, §2.3, §2.6, §2.7
18	Characterizing heterogeneity	Not reported	No subgroup analyses were performed
19	Characterizing distributional effects	Not reported	Distributional effects were not modelled

No.	Item	Reported	Location
20	Characterizing uncertainty	Yes	§2.7; Supplement §9
21	Approach to engagement with patients and others affected by the study	Not applicable	No patient, public or stakeholder engagement was undertaken
<i>Results</i>			
22	Study parameters	Yes	§2.2–2.5; Supplement §7 (complete parameter table)
23	Summary of main results	Yes	§3.1, §3.2 (Tables 2–3)
24	Effect of uncertainty	Yes	§3.4–3.6 (DSA, PSA, structural scenarios, price thresholds); Supplement §9–10
25	Effect of engagement with patients and others affected by the study	Not applicable	No engagement was undertaken
<i>Discussion</i>			
26	Study findings, limitations, generalizability, and current knowledge	Yes	§4.1, §4.5, §4.6
<i>Other relevant information</i>			
27	Source of funding	Yes	Separate title-page file
28	Conflicts of interest	Yes	Separate title-page file

Reported, Not applicable and Not reported are used as directed by the CHEERS 2022 statement [11]. Locations refer to the manuscript unless a supplement section is named. No analysis plan was registered or published for this evaluation, no subgroup or distributional analyses were undertaken, and no patients or stakeholders were engaged, as this is a model-based evaluation of published aggregate trial data.

References

- [1] A. G. Bensimon, Z. Y. Zhou, M. Jenkins, et al. Cost-effectiveness of pembrolizumab for the adjuvant treatment of resected high-risk stage iii melanoma in the united states. *Journal of Medical Economics*, 22(10):981–993, 2019.
- [2] A. D. Tran, G. B. Fogarty, A. K. Nowak, D. Espinoza, N. Rowbotham, M. R. Stockler, and R. L. Morton. A systematic review and meta-analysis of utility estimates in melanoma. *British Journal of Dermatology*, 178(2):384–393, 2018. doi: 10.1111/bjd.16098.
- [3] S. Zhang, A. G. Bensimon, R. Xu, et al. Cost-effectiveness analysis of pembrolizumab as an adjuvant treatment of resected stage iib or iic melanoma in the united states. *Advances in Therapy*, 40:3038–3055, 2023.
- [4] A. Khattak, M. S. Carlino, T. Meniawy, G. Ansstas, M. H. Taylor, K. B. Kim, et al. Intismeran autogene plus pembrolizumab versus pembrolizumab alone in high-risk resected melanoma: 5-year update of the randomized phase Iib KEYNOTE-942 study. *Journal of Clinical Oncology*, 44(26):2482–2489, 2026. doi: 10.1200/JCO-26-00835.
- [5] J. S. Weber, M. S. Carlino, A. Khattak, et al. Individualised neoantigen therapy mrna-4157 (v940) plus pembrolizumab versus pembrolizumab monotherapy in resected melanoma (keynote-942): a randomised, phase 2b study. *The Lancet*, 403(10421):632–644, 2024.
- [6] P. A. Ascierto, M. Del Vecchio, B. Merelli, et al. Nivolumab for resected stage iii or iv melanoma at 9 years. *New England Journal of Medicine*, 394(4):333–342, 2026. doi: 10.1056/NEJMoa2504966.
- [7] A. M. M. Eggermont, M. Kicinski, C. U. Blank, et al. Seven-year analysis of adjuvant pembrolizumab versus placebo in stage iii melanoma in the eortc1325 / keynote-054 trial. *European Journal of Cancer*, 211:114327, 2024. doi: 10.1016/j.ejca.2024.114327.
- [8] P. A. Ascierto, M. Del Vecchio, M. Mandalá, et al. Adjuvant nivolumab versus ipilimumab in resected stage iii/iv melanoma: 5-year efficacy and biomarker results from checkmate 238. *Clinical Cancer Research*, 29(17):3352–3361, 2023.
- [9] G. D. Sanders, P. J. Neumann, A. Basu, et al. Recommendations for conduct, methodological practices, and reporting of cost-effectiveness analyses: Second panel on cost-effectiveness in health and medicine. *JAMA*, 316(10):1093–1103, 2016.
- [10] P. J. Neumann, J. T. Cohen, and M. C. Weinstein. Updating cost-effectiveness — the curious resilience of the \$50,000-per-qaly threshold. *New England Journal of Medicine*, 371(9):796–797, 2014.
- [11] D. Husereau, M. Drummond, F. Augustovski, et al. Consolidated health economic evaluation reporting standards 2022 (cheers 2022) statement: updated reporting guidance for health economic evaluations. *Value in Health*, 25(1):3–9, 2022.